

Numerical Production Contract for an Adaptive Chiral-Defect CFT

Asadullah Bhuiyan¹

¹*Department of Physics, Cornell University, Ithaca, New York 14853, USA*

(Dated: August 16, 2026)

This document is a preregistered numerical specification for the proposed PRX Quantum manuscript on a postselection-free adaptive class-A fermion circuit. It contains no numerical result and licenses no present-tense physical claim. Existing repository code and artifacts are implementation references only. The contract fixes the physical ensemble, system-size sequences, sample counts, burn-in logic, saved data, estimator order, controls, uncertainty analysis, acceptance tests, claim downgrades, and five-figure architecture. Every GPU circuit trajectory must be generated by the canonical `classA_U1FGTN_gpu.run_markov_circuit` entry point on an A100-class 40 GB runtime. The central acceptance standard is deliberately conjunctive: quantitative revalidation of the inherited BPJ bulk, a $c = 1$ and level-one $U(1)$ wall state, physical chiral response, an integer twist spectral flow, a wall-localized tangent sector controlling purification, and universal information recoverable from trajectory-resolved records must all survive thermodynamic, transverse, temporal, and protocol controls.

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I. SCOPE AND EVIDENTIARY RULES

All stochastic production campaigns below retain status PLANNED. The deterministic B0 review calibration is now completed and is recorded explicitly in its subsection. No saved plot, notebook output, or previous fit is admissible as production evidence. Repository artifacts may be cited only in a field labelled “legacy comparison,” and only when the observable and protocol have been checked at the path level. Such a comparison is never pooled with production data.

The physical object is the Born ensemble of adaptive trajectories. For a record $\xi = (m_1, \dots, m_T)$,

$$K_\xi = K_{m_T|\xi_{<T}} \cdots K_{m_1}, \quad p_\xi = \text{Tr}(K_\xi \rho_0 K_\xi^\dagger), \quad \rho_\xi = K_\xi \rho_0 K_\xi^\dagger / p_\xi. \quad (1)$$

Perfect correction samples every m_t from its conditional Born probability and then applies the prescribed feedback. Forced postselection is a control. The mean channel is a second, inequivalent control. For every nonlinear observable,

$$\mathbb{E}_\xi[O(C_\xi)] \neq O(\mathbb{E}_\xi[C_\xi]) \quad (2)$$

in general. The estimator must therefore be evaluated per trajectory before equal-weight averaging. Since the trajectories are Born sampled, no second probability weight is applied.

The manuscript is ready for a PRX Quantum submission only if the five gates in Table I pass. A campaign may produce an interesting result while failing its predeclared gate; in that case the corresponding weaker wording, not a modified estimator, is used.

TABLE I. Conjunctive manuscript gates. “Pass” means that every campaign-specific finite-size, transverse, temporal, and control test has also passed.

gate	required result	failure consequence
G1	fresh-ancilla $n_{\text{shell}} = 1$ realization reproduces the quantized BPJ bulk phase	remove the validated-topological-platform claim
G2	state data give $c_{\text{edge}} = 1$, $U(1)$ level $k = 1$, and the expected correlator	say “critical wall,” not “chiral defect CFT”
G3	total Born response is chiral and frozen-record flux has integer flow	do not claim operational chirality or a branch-wise integer wall index
G4	a wall-localized tangent gap closes and controls max-mix purification	retain tangent construction only as a finite-time diagnostic
G5	replica-calibrated record self-information and charge sectors reproduce universal defect data	remove the record-only universality claim

II. PRX QUANTUM THESIS AND INHERITED BPJ BASELINE

The numerical program is not designed to re-establish the premises already supplied by the Bhuiyan–Pan–Jian (BPJ) construction [1]. BPJ already provides the mEO/sTM setting, a postselection-free finite-range class-A adaptive instrument, Chern trajectory ensembles, programmable dynamical domain walls, finite-range parallelization, an ancilla–fSWAP implementation route, and quantum-simulation prospects on encoded-qubit or native-fermion hardware. It also reports coherent-noise robustness and distinct trajectory and averaged thresholds. Accordingly, this manuscript shall not advertise postselection freedom, ancilla feedback, Gaussian simulability, finite-range state preparation, or the mere existence of a domain-wall slow mode as its standalone novelty.

The proposed advance is the conjunction of results absent from that baseline. The perfect-correction instrument should program a *universal chiral defect* across typical retained Born trajectories: each trajectory may occupy a different definite system-charge sector, but the ensemble shares $c_{\text{edge}} = 1$ and level-one $U(1)$ wall data, its sector wandering is localized at the walls, its response and flux flow are oriented, and its slow purification is controlled by a singular conditional tangent sector. The strongest version further recovers the defect data from the classical record and shows precisely what survives or disappears in the unconditional channel. This is the unique intersection to which every headline panel must contribute.

The quantum-simulation proposition is therefore operational rather than computational. It is not a quantum-advantage claim, nor a claim that an adaptive platform can prepare a Chern state for the first time. It is a protocol for realizing and interrogating a programmable topological defect ensemble without rejecting records, with a hierarchy of record-only, number-measurement, covariance-tomography, and replica observables. A result that cannot be placed in this hierarchy or tied to one of Gates G1–G5 is supplemental or deferred even if it is numerically interesting.

III. CANONICAL PROTOCOL AND FIXED PRODUCTION CHOICES

Every GPU script or notebook that simulates the circuit shall configure and call `classA_U1FGTN_gpu.run_markov_circuit`. It shall not reproduce the cycle loop or call private local-update helpers. New dynamics functionality belongs in the canonical class. Every output manifest records this entry point explicitly.

The base physical protocol is

$$n_{\text{shell}} = 1, \quad \alpha_{\text{top}} = 1, \quad \alpha_{\text{triv}} = 30, \quad \text{perfect_correction}=\text{True}, \quad \text{postselect}=\text{False}. \quad (3)$$

The production implementation is the ordered serial instrument implemented by the canonical dynamics engine. A fresh random permutation of the active centers is drawn for each trajectory and cycle and stored in metadata; the four controller channels at a center are applied in the fixed declared engine order. Raster and reverse-raster words are order-artifact controls, not alternative labels for the same microscopic circuit. Campaign V0 validates exact schedule replay and Campaign V1 checks that the headline conclusions are not created by a preferred serial scan direction. The candidate quasi-one-dimensional sequence is

$$N_x = 20, \quad N_y \in \{24, 32, 40, 48, 64, 80\}, \quad (4)$$

but $N_x = 20$ is not accepted until Campaign W1 passes. Every stochastic parameter point uses a base batch of $S = 100$ independently burned-in trajectories. If a predeclared confidence interval is too broad to decide its gate, launch a

second complete $S = 100$ batch for *every* member of the affected comparison family, with a new root seed. Never add samples only to an inconvenient point. The first and second batches remain identified in metadata and are combined only after a heterogeneity check.

All topology, tangent, twist, and fit inputs use complex128 or float64. Figure defaults are 3.375 inches per single-column panel, 300 dpi, CMU Sans Serif, and at least 8 pt text.

IV. QUANTUM-SIMULATION IMPLEMENTATION AND OPERATIONAL OBSERVABLES

The primary model is the fresh-ancilla, perfect-correction idealization of the BPJ adaptive circuit. For a controller mode χ_j with target occupation $s_j \in \{0, 1\}$, measure n_{χ_j} and retain the Born outcome m_j . If $m_j \neq s_j$, prepare a fresh local ancilla mode in occupation s_j , fSWAP it with χ_j , and trace out or reset the outgoing ancilla. Thus the system charge transfer is

$$Y_j = s_j - m_j, \quad (5)$$

and every outcome is a valid retained run. No success flag, trajectory rejection, or reweighting is permitted in the primary data. The older dynamical-bath implementation, where correction succeeds only when a reused ancilla happens to have the needed occupation, is an imperfect-feedback control rather than the definition of perfect correction.

For $n_{\text{shell}} = 1$, an instrument block may be compiled by a number-conserving Gaussian basis rotation on the 3×3 -cell patch that maps χ_j to a readout mode, followed by occupation readout, classical feedforward, target-occupation ancilla preparation, fSWAP, and the required inverse basis rotation. The hardware report shall give the two-mode-gate count and depth of this compilation as d_{inst} , the measurement and feedforward latency, the number of concurrently live ancillas, reset fidelity, and the colored-sweep bound in Eq. (C6). The same ancilla pool can be recycled between color/channel blocks. Fermionic locality does not automatically imply qubit locality in two dimensions: a qubit implementation must state its Jordan–Wigner, Bravyi–Kitaev, swap-network, or auxiliary-gauge encoding and include that mapping overhead. A native fermionic processor may use the fermionic compilation directly.

Because controllers within each conflict-graph color have disjoint support, their local instruments commute and admit simultaneous execution on a quantum simulator. A numerical evaluation of such a color block may nevertheless apply its instruments sequentially, which samples the identical joint Born law while avoiding a separate correlated multi-mode sampler. This optional finite-color compilation is a quantum-simulation resource advantage, not a requirement of the production numerics. Reordering noncommuting color blocks relative to the production random-serial word would define a different microscopic schedule and is not claimed to leave finite-size trajectories unchanged.

The operational hierarchy is:

1. the measurement/correction record, activity, transfer FCS, and total system-charge history require no state tomography;
2. boundary charge profiles, Eq. (27), and the real-time response require repeated occupation measurements and local source control;
3. the one-body covariance, local marker, and occupation spectrum require a correlation-matrix tomography protocol;
4. Rényi entropies require replicas, randomized measurements, or an equivalent interferometric protocol, while modular-Hamiltonian evolution may remain a trajectory-reconstruction diagnostic rather than a direct hardware primitive.

The simulation package shall attach shot-complexity and calibration estimates to each observable proposed as operational. Since Gaussian circuits are classically simulable, the manuscript does not claim computational quantum advantage. The quantum simulation value is controlled realization of a programmable topological defect and access to typical conditioned states and their associated trajectory records, without conditioning on an exponentially unlikely favorable record.

This implementation inherits rather than supersedes the BPJ hardware proposal. Its concrete new simplification is that a fresh ancilla is prepared in the target occupation s_j for each wrong branch, so the correction is deterministic once the Born outcome is known; no ancillary redistribution layer or finite bath-occupation success probability is part of the default model. Production resource tables must compare this fresh-ancilla cost with the reused-bath BPJ primitive instead of presenting the whole measurement–fSWAP idea as new.

The ensemble language must also remain precise. Starting from a definite total charge, each conditioned trajectory lies in a definite system-charge sector after every recorded gain/loss event, while the ensemble spans different sectors.

The intended result is therefore a Born-sampled ensemble of boundary states with shared infrared data, not one deterministically prepared ket and not an individual charge-indefinite pure state. Campaign S1 must demonstrate that the trajectory-to-trajectory charge wandering is localized at the walls and separately extract the within-trajectory current-algebra coefficient.

V. REPRODUCIBILITY, SHARDING, AND OUTPUT SCHEMA

Each campaign has an immutable JSON manifest containing a schema version, git commit, canonical entry point, environment versions, device, dtype, complete geometry and protocol, observer definitions, root seed, the ordered list of sample seeds, shard allocation, and analysis version. A counter-based random stream keyed by sample, cycle, elementary update, and stochastic role is preferred. Otherwise the exact generator state is checkpointed.

A production directory contains

$$\text{manifest.json, shards/, merged/, analysis/, figures/, logs/}. \quad (6)$$

Sample indices are assigned to shards before execution. The merger rejects duplicate or missing indices and any manifest hash mismatch. Atomic checkpoints are written after each observation block and contain the covariance, record cursor, RNG state, accumulated $-\log p$, observer accumulators, and, when present, the tangent frame and QR log sums. Resume, serial, and sharded runs must agree with an uninterrupted small-system run within declared tolerances.

Required raw products include per-trajectory scalars, not only ensemble averages; bit-packed records; conditional log probabilities or lossless block sums; equilibrium covariance snapshots in a chunked format; numerical residuals; censor/null masks; and a completion table. Analysis writes figure-ready CSV/NPZ files and a fit JSON containing the input data hash, fit window, covariance model, bootstrap distribution, and sensitivity fits. Plotting scripts do not perform unrecorded fits.

VI. MASTER TRAJECTORY FAMILIES AND REUSE RULE

A new independent trajectory family is required only when at least one component of the following key changes:

$$\mathfrak{K} = (N_x, N_y, \text{controller construction and parameters, initial ensemble, serial schedule law, feedback/noise law, } n_{\text{shell}}). \quad (7)$$

Changing the observer, fit, source position, analysis window, or deterministic replay does not define a new Born family. All campaigns sharing $\mathfrak{K}$ must consume one common set of independently seeded trajectories rather than launch campaign-specific copies. One trajectory remains the independent sampling unit even when many cycle snapshots, periodic origins, subsystem positions, sources, or twists are derived from it.

For the primary pure-state wall and its matched wall-off control, run one master family per accepted size. After the common burn-in, retain at least $32N_y$ cycles so the first $2N_y$ supplies stationary state observables, the tangent observer can discard its N_y alignment interval and accumulate through $16N_y$, and R1 can use nested record blocks out to aspect ratio 32. The same family supplies S1, T1, B2, the stationary states for H1, the unperturbed parents and time origins for H2, the frozen records for H3, R1, and the Born comparison in M1. Response continuations in H2 and twist replays in H3 are derived jobs attached to a parent trajectory ID; they are not new independent samples.

For max-mix initialization, run one separate master family per wall/control geometry and size from $t = 0$ through at least $16N_y$. It supplies P2 and the max-mix portion of S1. Uniform bulk, secondary-wall, changed- α , changed-noise, imperfect-feedback, postselected, changed- n_{shell} , and changed serial-word controls have different keys and therefore cannot be manufactured from a baseline trajectory. B0 and the B1 pilot are deterministic or small-pilot calculations; H1 modular evolution, H3 fixed-record flux evolution, and the exact mean maps in M1 are deterministic descendants of saved states or records.

Each pure-state master trajectory saves:

1. its sample seed, initialization hash, controller hash, complete ordered site/channel word, bit-packed outcomes, per-event conditional log probability, X_j , Y_j , and the system charge after every event or a losslessly reconstructible block representation;
2. the initial covariance, burn-in endpoint, tangent-alignment endpoint, the three S1 stationary snapshots, all H2 source-time snapshots, final covariance, RNG/checkpoint state, and replay hashes, but not an unneeded dense covariance at every cycle;

3. per trajectory, snapshot, and Rényi index, the y_0 -averaged full-strip entropy curve, selected y_0 -resolved validation curves, unnormalized wall-window contour sums, subsystem spectra, charge-susceptibility spectra $q_a = \nu_a(1 - \nu_a)$, matched entropy–variance residual curves, interval-charge cumulants, density and charge profiles, structure factors, the full fixed-slice array $C_{G,\xi}(x, r_y)$, its wall/bulk derived curves, topology and purity diagnostics, and the simultaneous $N_{\mathcal{D},\xi}$ needed for sector-resolved fits;
4. tangent QR log increments, null masks, aligned low-subspace projectors and densities, block- R factors at declared checkpoints, and the initial frame seed; and
5. observer residuals, censor masks, completion status, and enough accumulated score data or replay information to reconstruct the H2 conditional and score terms.

The max-mix master additionally saves every early-cycle scalar, logarithmically spaced late covariances and contours, the complete start-to-time record, and the reduced Choi/tangent objects required by Eq. (38). Scan-only families may save the same per-trajectory fitted scalars and a smaller declared set of covariance checkpoints; they must not discard the sample axis or save only ensemble means.

A. Minimum gate-first execution matrix

The minimum below assumes that B0 accepts $N_x^* = 20$, one base batch of $S = 100$ resolves every gate, the explicit interface is the primary wall, and the support-terminated construction is the secondary wall. It is a count of independently sampled trajectory families, not deterministic exact calculations, observer passes, response continuations, or fixed-record replays. The fixed core is

family block	$S = 100$ families trajectories	
P1 lean bulk baseline	20	2000
primary pure wall and matched control, six sizes	12	1200
primary max-mix wall and matched control, six sizes	12	1200
secondary pure wall and matched control, three sizes	6	600
W1 transverse-convergence additions	18	1800
V1 raster and reverse-raster controls	2	200
core through G1–G4	70	7000
R1 extra wall/control families after the $L = 20$ reuse	5	500
core through G1–G5	75	7500
S2 nonbaseline α_{in} points	32	3200
fixed campaign including S2	107	10700

Here P1 uses pure topological and trivial states at all six square sizes, max-mix states only at $L = 20, 28$, and the $n_{\text{shell}} = 2$ check only for the pure topological and trivial states at $L = 20, 28$, giving $12 + 4 + 4 = 20$ families. The W1 count uses four protocols—two walls and their separate matched controls—at $N_x = 20, 24, 28$ and $N_y = 48, 80$. Of these 24 points, six are already master points, so only 18 are new. The S2 count is $9 \times 4 = 36$ points minus the four baseline $\alpha_{\text{in}} = 1$ families already present. R1 adds the primary wall and control at $N_y = 16, 20, 28$, except that the $N_x = N_y = 20$ matched all-trivial control reuses the pure $L = 20$ P1 trivial family by extending its saved stationary record. This reuse is allowed only after their controller, active-channel, schedule-law, and initialization hashes agree; otherwise the count increases by one.

All other fixed campaigns reuse those samples. B0 and the mean maps in M1 are deterministic. T1 is an observer/replay of the pure master record. H1 evolves saved stationary covariances, H2 attaches response continuations to saved parent trajectories, and H3 replays a saved record around the twist circle. B2 consumes the three common secondary-wall sizes. The two V1 ordered-word families are retained long enough to also supply the ordered-word H2 and M1 controls. The B1 train/test pilot and V0 exact checks are small validation jobs and are reported separately from the $S = 100$ production count.

The open perturbation grids add

$$N_{100} = 107 + 3n_{\text{imp}} + 17n_{\text{noise}}, \quad (8)$$

where n_{imp} is the number of nonperfect stochastic feedback points in M2 and n_{noise} is the number of nonzero noise values in M3. The deterministic postselected branch adds no stochastic family. Each added M3 noise value contributes

five uniform square sizes and two wall geometries at six circumferences. Equation (8) excludes any escalation triggered by a failed gate. A second base batch adds one further $S = 100$ family for every member of the affected comparison block, never only for one selected point.

This minimum is valid only if every master run writes the complete save contract above. In particular the manifest and record must be sufficient to replay the exact ordered instrument; the three stationary covariances must support all state, modular, twist, and correlator analyses; the event stream must retain conditional log probabilities and signed charge transfers for B2 and R1; the tangent QR stream must span the full T1 window; and H2 must retain its declared parent snapshots, score accumulators, and source-time metadata. If one of these products is omitted, reconstructing it later may require a new independent family and invalidates the minimum count.

VII. VALIDATION LADDER

A. Campaign V0: engine and observer validation

a. Claim and status. Status PLANNED. The implementation reproduces exact small-system probabilities, covariances, derivatives, and observables and is invariant under resume and sharding.

b. Dependencies and run matrix. None. Use 4×6 and 6×8 systems, four deterministic seeds, three to ten cycles, CPU dense references, and GPU batch sizes one and four.

c. Initialization, burn-in, and observation. No burn-in is needed. Compare every elementary update and every cycle boundary.

d. Raw saves and estimator. Save branch probabilities, records, covariances, spectra, charge changes, tangent actions, and RNG states. Every CPU/GPU comparison passes an explicit common wall mask, `DW_loc`, active-index list, and controller-projector hash; it never relies on engine defaults, because the current CPU and GPU defaults choose different wall fractions. Enumerate all records on the smallest system and check $\sum_{\xi} p_{\xi} = 1$. Compare analytic tangent-vector products with centered finite differences over at least five perturbation sizes.

e. Controls, statistics, and gate. Require covariance Hermiticity and spectral bounds, exact charge bookkeeping, CPU–GPU agreement only after the wall/projector hashes match, uninterrupted–resume agreement, serial–shard agreement, and 2π twist closure. Tolerances must be written in the manifest and justified from float64 roundoff. Any failure blocks all production.

f. Downgrade, figure, sharding, and legacy. There is no scientific downgrade and no main-text figure. No legacy comparison is needed.

B. Campaign V1: serial schedule and order-artifact check

a. Claim and status. Status PLANNED. The production model is the random serial ordered instrument. This campaign validates schedule generation and replay and checks that its main conclusions are not artifacts of a directed raster word. Noncommuting serial words are not expected to produce identical finite-size states.

b. Dependencies and run matrix. Depends on V0. Verify random serial permutations, raster order, and reverse-raster order on the V0 sizes. At $N_x = 20, N_y = 48$, compare the production random serial word with raster and reverse-raster controls using $S = 100$ independent trajectories per word. Escalate to additional sizes only if a headline conclusion changes at the representative size. The random family is the primary master already counted; retain the two ordered-word controls through the common stationary and response windows so they can also supply H2 and M1.

c. Initialization, burn-in, and observation. Use the same declared product-state ensemble for all words. Determine burn-in separately for each word; do not recycle one word’s burned-in covariance as another’s initial state.

d. Raw saves and estimator. Save every cycle’s ordered site list and seed, per-channel update counts, physical instrument-block count, and the tested observables. V0 must verify one-and-only-one visit to every active center and channel, periodic wrapping, exact replay, CPU–GPU agreement, and checkpoint restart.

e. Controls, statistics, and gate. Require stable bulk topology, wall localization, entropy and interval-charge coefficients, and chirality sign under raster reversal within declared finite-size uncertainty. This is an order-artifact check rather than a demand for pointwise covariance agreement. A schedule-sensitive nonuniversal amplitude is reported as such.

f. Downgrade, figure, sharding, and legacy. The random serial word remains the production protocol. If a headline conclusion changes under raster reversal, report that schedule dependence and do not call the conclusion an order-independent infrared property. This is a supplemental validation figure. Existing raster results remain parameter pilots and are not relabeled as random-serial data.

VIII. EXACT DOMAIN-WALL CALIBRATION

A. Campaign B0: deterministic analysis-pipeline benchmark

a. Claim and status. Status COMPLETED REVIEW CALIBRATION. The common analysis pipeline recovers the known chiral wall behavior of an exact class-A domain-wall Hamiltonian. This calibrates geometry and fit windows; it is not evidence about the adaptive circuit.

b. Dependencies and run matrix. Depends on V0. Use `classA_U1FGTN_domain_wall_hamiltonian` and `G_CI_domain_wall`. At $N_y = 48$, scan $N_x = 12, 16, 20, 24, 28, 32$. At the selected width scan the sequence in Eq. (4). Run both the exact coupled spatial-mass interface and `triv_region_local_mode`, a decoupled-exterior hard analogue. Neither is assumed identical to controller `dw_truncation`.

c. Initialization, burn-in, and observation. There is no stochastic burn-in. Enforce exact half filling. Resolve zero-energy crossings with a small boundary twist and continuous overlap tracking; test both twist signs and the zero-twist limit. Never fill states solely by $E < -\epsilon$.

d. Raw saves and estimator. Save Hamiltonians or their reproducible construction metadata, low spectra and vectors, wall weights, entanglement spectra, the complete fixed- x correlator tensor for every x and $1 \leq r_y \leq \lfloor N_y/2 \rfloor$, modular responses, density responses, and twist-overlap matrices. Recompute the wall, exterior-neighbor, topological-bulk, trivial-bulk, wall-window, and full- x correlation curves and their fit tables at every benchmark size; no archived curve or fitted exponent is imported. Apply without alteration the estimators later used in S1 and the three chirality campaigns.

e. Controls, statistics, and gate. Extract edge velocity, transverse localization length, finite-width hybridization, $c_{\text{edge}} = 1$ entropy coefficients in the one- and two-wall geometries, an r^{-2} squared correlator, opposite modular/physical velocities, and unit twist flow. Select N_x only when the hybridization mass satisfies

$$m(w) \ll \frac{2\pi v}{N_y}. \quad (9)$$

Failure means the analysis pipeline or geometry must be repaired before stochastic runs.

f. Downgrade, figure, sharding, and legacy. The calibration becomes Fig. S1, not a primary result. Deterministic sizes may be sharded independently. Path-audited visual references are `exact_DW_benchmark_notes/DW_GS_Benchmark_Figs/energy_spec.fig.png`, `integrated_contour_vs_log_chord.png`, `square_correlation_profile.y.png`, and `mod_current.png` in the same directory. They are not production inputs.

g. Current audited campaign. The authoritative version-2 calculation is campaign `experiment_review/b0_exact_domain_wall/results/20260816_191957/`; all entries in its machine-readable acceptance ledger pass. It preserves but supersedes version-1 campaign `20260816_184251` for interpretation. The first run used a normalized packet at the center of the retained half-cylinder and fitted an absolute center of mass. That estimator mixed oppositely directed modular components and returned small same-sign slopes. It also selected $N_x = 12$ from the hybridization mass alone, even though the coupled local correlator and corrected modular response remain underconverged at $N_x = 12, 16$. The discrepancy is classified as an estimator mismatch together with a finite-width convergence effect, not as a failure of modular charge spreading.

Version 2 retains the mass gate but requires every single-geometry calibration to pass at both the accepted width and the next larger scanned width. This selects $N_x^* = 20$ because $N_x = 20$ and 24 pass whereas 12 and 16 fail the coupled local checks. For modular flow it evolves the $y = 0$ and $y = N_y/2 - 1$ sources separately at each physical wall and fits

$$D_w(t) = \frac{1}{2} [\bar{y}_{w,0}(t) + \bar{y}_{w,L-1}(t) - (L-1)], \quad L = N_y/2, \quad (10)$$

on $0.1 \leq t \leq 2.0$. One- and three-column sources and three alternate time windows give the same sign. Table II records the accepted-width values; modular slopes use modular-time units and are not compared in magnitude with physical velocities.

h. Legacy comparison. **Assessment: promising and already a substantial single-geometry calibration, but not a B0 pass.** A repository-wide audit finds quantitative exact-state entropy and correlator fits for the coupled interface, corresponding fit tables for `triv_region_local_mode`, and raw exact modular-evolution arrays. The strongest path-audited results are summarized in Table III. Here m_{full} is the two-wall full-width von Neumann log-chord slope, m_{wall} is the unnormalized single-wall contour-window slope, and β is defined by $C_G \sim \ell_y^{-\beta}$.

The principal defect is the zero-mode prescription. The archived periodic spectrum has two modes at numerical zero, whereas `G_CI_domain_wall` occupies only $E < -\text{tol}$. Reconstructing the archived 20×40 , $x = 6, \dots, 14$ geometry with that audited rule selects 799 modes for the coupled interface rather than the required 800. For the hard-exterior

TABLE II. Audited version-2 B0 results at $N_x = 20, N_y = 48$. The complete campaign also contains the accepted-width sequence $N_y = 24, 32, 40, 48, 64, 80$, all raw packet fields, and the 33/65/129-point twist controls.

construction	representative values	audited implication
Coupled spatial-mass interface	$c_1 = 1.0001, \beta = 2.07447$; $v_{\text{phys}} = (-0.87203, +0.87203)$; $v_{\text{mod}} = (-2.93045, +4.53563)$	The three-column modular source retains at least 0.3245 charge in the primary fit window, all sensitivity signs agree, and the norm drift is 3.3×10^{-15} . The wall twist flows are $(-1, +1)$.
Hard-exterior analogue	$c_1 = 1.0001, \beta = 2.07437$; $v_{\text{phys}} = (-0.87203, +0.87203)$; $v_{\text{mod}} = (-2.59985, +4.40578)$	The minimum modular retention is 0.3349, all sensitivity signs agree, and the norm drift is 4.0×10^{-15} . The wall twist flows are again $(-1, +1)$.
Legacy-protocol reproduction at 20×40	Aggregate initial charges 8 and 24; $0 \leq t \leq 40$ with $\Delta t = 0.05$	The complete aggregate $N_{xy}(t)$ fields are saved with norm drift 1.4×10^{-14} , while the separate-source data make handedness identifiable.

TABLE III. Legacy B0 evidence. These deterministic results motivate the production fit windows but are not pooled with the preregistered campaign.

artifact and coverage	strongest quantitative indication	implication and limitation
Coupled spatial-mass interface in <code>notebooks/exact_DW_GS_benchmark.ipynb</code> and <code>exact_DW_benchmark_notes/</code> , 20×40 , two walls at $x = 6, 14$	$m_{\text{full}} = 0.343924$ ($3m_{\text{full}} = 1.03177, R^2 = 0.999080$); $m_{\text{wall}} = 0.168600$ – 0.168642 ($R^2 \simeq 0.999875$); $\beta_{\text{wall}} = 2.024641$ ($R^2 = 0.999835$) on $r_y = 3, \dots, 15$. The saved spectrum shows two low-energy crossing branches, while the marker and modular current maps are wall localized.	Strong evidence that the existing geometry and estimators recover the expected $c = 1$ and r^{-2} structure at one size. It is a two-wall calculation only and has no finite-width or longitudinal-size extrapolation.
Exact fit tables in <code>notebooks/flattened_hamiltonian_analysis/data/slope_fit_df.csv</code> and <code>notebooks/flattened_hamiltonian_analysis/data/cavg_fit_df.csv</code> , 20×40	For <code>triv_region_local_mode</code> , $m_{\text{full}} = 0.343923$, $m_{\text{wall}} = 0.168600$ – 0.168642 , and $\beta_{\text{wall}} = 2.024671$ with $R^2 = 0.999835$. The coupled and hard-exterior values differ by less than 2×10^{-6} in the entropy slope and 3.1×10^{-5} in β .	Very promising coupled-versus-hard agreement, but only for the same 20×40 geometry. The table preserves fits rather than Hamiltonians, eigenvectors, or complete spectral data.
Current-normalization correlation archive in <code>colab_small_system_testing/analysis_outputs/pure_state_square_correlations_cpu/</code> , 16×30 and 16×40	Complete scalar $C_G(x, r_y)$ curves are saved for every x and $0 \leq r_y \leq N_y/2$. The wall exponents are respectively $\beta = 2.042461$ and 2.040144 , with $R^2 = 0.999742$ and 0.999753 on $r_y = 3, \dots, 10$; the full- x averages give $\beta = 2.11628$ and 2.10899 .	Confirms modest N_y stability using the normalization of Eq. (24) and supplies reusable raw curves and fit rows. It still lacks the B0 N_x scan and the preregistered $r_y = 3, \dots, 15$ model comparison.
Exact modular data in <code>notebooks/modular_charge_spreading/exact_dw_gs/</code> and the static map <code>exact_DW_benchmark_notes/DW_GS_Benchmark_Figs/mod_current.png</code> , 20×40	The static J_y map has opposite signs at the two interfaces. Two saved exact evolutions contain the full $N_{xy}(t)$ array for $0 \leq t \leq 40$ with total-charge drift below 4.4×10^{-7} .	Provides raw material for a modular-drift calibration, but no wall velocity was fitted; there is no corresponding physical density-response velocity or orientation-reversal control.

Hamiltonian it selects 1239 modes because the decoupled exterior orbitals lie at energy -1 . Thus the excellent legacy slopes do not yet constitute the exact-half-filled, twist-tracked benchmark demanded above. The artifacts also predate the present domain-wall convention: they use walls at $x = 6, 14$, and some notebooks use the historical reversed `alpha_1/alpha_2` naming, whereas the current source uses a wider slab and explicit topological/trivial aliases.

At the time of the legacy audit, no completed artifact was found for the required $N_y = 48$ scan over $N_x = 12, 16, 20, 24, 28, 32$, the accepted-width scan over Eq. (4), a one-wall entropy geometry, edge velocity, transverse localization length, hybridization mass, physical density response, twist-overlap tracking, or unit twist flow. A source script for an exact entropy size sweep exists at `scripts/run_markov_size_sweep_integrated_contour_fit_equilibrium.py`, but its declared output is absent and the script saves no numerical fit table, so it is not counted as a completed legacy run.

B. Pilot B1: signed controller-frame incompatibility diagnostic

a. Claim and status. Status PLANNED exploratory pilot. This calculation asks only whether a static fixed-charge incompatibility diagnostic has enough quantitative predictive value to merit later expansion. It is not a manuscript gate, a model of the ordered trajectory dynamics, or evidence for topology, a wall CFT, or slow conditioned dynamics.

b. Formal object. For controller targets $s_j \in \{0, 1\}$ and normalized rank-one projectors P_j , define

$$H_{\text{frame}} = \sum_j (1 - 2s_j) P_j. \quad (11)$$

The static target cost over rank- N Slater projectors is

$$\mathcal{F}[C] = \sum_j s_j + \text{Tr}(H_{\text{frame}} C), \quad C^2 = C, \quad \text{Tr } C = N. \quad (12)$$

Ky Fan minimization therefore gives

$$C_\star = \sum_{a=1}^N |u_a\rangle\langle u_a|, \quad F_\star = \sum_j s_j + \sum_{a=1}^N \epsilon_a^\dagger(H_{\text{frame}}). \quad (13)$$

This evaluates all constraints on one common covariance. Perfect correction instead changes both the covariance and, on wrong branches, the system charge after each ordered event. Consequently $F_\star(N)$ is neither a lower bound nor a dynamical variational principle for the realized activity. The object called the flattened Hamiltonian in the legacy notebooks is therefore used here only as a signed controller-frame diagnostic; its spectrum is not generally ± 1 , and it is neither the conditioned tangent operator nor the mean channel.

c. Dependencies and run matrix. Depends on V0. Use $N_x = 20$, $N_y = 24, 40$, and $n_{\text{shell}} = 1$ for the two wall constructions and their geometry-matched wall-off controls. Evaluate nominal half filling and the charge sectors encountered in a four-trajectory stationary pilot. Do not launch a width, support, or parameter sweep at this stage.

d. Raw saves and estimator. Save the frame hash, spectrum of H_{frame} , $F_\star(N)$, selection gaps, and the charge-resolved transverse residual profiles. Verify Eqs. (12) and (13) by direct residual summation. Compare the residual profiles, without shifting or refitting their spatial origins, with the wrong-outcome activity measured on a disjoint four-trajectory test set. Do not run the CFT, topology, twist, or chirality pipeline on $C_{\star, N}$.

e. Controls, statistics, and gate. Nonzero incompatibility is generic for nonorthogonal or overcomplete constraints and is not called topological frustration. If a selection gap closes, report the allowed degenerate-subspace variation rather than choosing a favorable minimizer. The pilot is expanded only if its fixed, charge-resolved residual predicts more than generic wall colocalization: require an out-of-sample profile overlap and wall-to-bulk contrast that both improve substantially over the matched wall-off profile.

f. Downgrade, figure, sharding, and legacy. There is no main-text figure and no failure consequence. Unless the pilot passes its expansion criterion, stop the static analysis and mention at most the defining algebra in the Supplement. Available legacy inputs are `notebooks/flattened_hamiltonian_analysis/docs/flattened_hamiltonian_analysis.tex`, the data and figures beside it, and `topological_frustration_diagnostics/docs/numerical_diagnostics.tex`. They guide sizes and estimator debugging only. Legacy postselected Frobenius convergence does not validate the Born-sampled perfect-correction circuit.

C. Campaign B2: feedback activity and slow dynamics

a. Claim and status. Status PLANNED. This campaign tests the dynamical association

$$\text{Born feedback activity} \longleftrightarrow \text{wall-localized slow tangent and purification sectors}, \quad (14)$$

rather than treating several wall-centered plots as independent observations. Pilot B1 is an optional comparison and is not required for this campaign.

b. Exact branch relation. Immediately before channel j , the covariance is C_{j-} and the wrong-outcome indicator and signed system transfer are

$$X_j = |m_j - s_j|, \quad Y_j = s_j - m_j. \quad (15)$$

For the perfect-correction primitive,

$$\mathbb{E}[X_j | C_{j-}] = s_j + (1 - 2s_j) \text{Tr}(P_j C_{j-}), \quad |Y_j| = X_j \quad (16)$$

event by event. This identity, evaluated on the actual pre-event covariance, is the primary microscopic statement. No simultaneous-frame cost or static variational bound is assumed for the ordered sweep.

c. Dependencies and run matrix. Depends on the burn-in pilot, T1, and P2. Reuse the production serial schedule and controller hashes at the common secondary-wall sizes $N_y = 32, 48, 64$, $S = 100$, for both wall constructions and their separate geometry- and update-count-matched wall-off controllers. These are the S1/W1 master families, so B2 launches no independent production trajectories. P1 supplies the uniform bulk reference. Changed n_{shell} , support, or slab width is an escalation only if the primary wall-versus-control association is unresolved; such an escalation receives its own preregistered grid and is not part of the minimum count.

d. Raw saves and estimator. Save every X_j, Y_j , controller label, position, pre-branch probability, schedule position, and observation-block index. Measure stationary signed and absolute activity profiles, wall/interior rates, inter-event correlations, and the joint fluctuations of activity, total system charge, tangent gap, and residual max-mix entropy. Compare the activity profile directly with the globally normalized low-tangent-subspace density and purification residual using profile overlap, Spearman rank correlation, and independently fitted localization lengths. Condition the ordered within-cycle record on its instantaneous N_j - where useful. A B1 residual may be shown as an explicitly exploratory overlay only if its independent pilot criterion passed. Correlations across different times from one trajectory do not increase the trajectory sample count.

The empirical activity function

$$\theta_R(s, T) = T^{-1} \log \mathbb{E}_\xi [e^{-s \sum_{j \in R, T} X_j}] \quad (17)$$

may be reported with log-sum-exp stabilization and reweighting effective sample size. It is a finite-time annealed record statistic, not the replica-calibrated defect self-information in R1, not a trajectory-resolved CFT partition function, and not a tilted-channel eigenvalue unless the tilted operator is explicitly constructed and its long-time limit is converged.

e. Controls, statistics, and gate. Validate Eq. (16) branch by branch and the mean profile against the M1 deterministic covariance prediction. Require a stationary interface excess over the matched active-channel control. To support a dynamical association, the same parameter sweep must link increased activity to the closing tangent sector and delayed purification with a preregistered monotone or scaling relation. Wall colocalization alone does not pass, and causation is not inferred from correlation.

f. Downgrade, figure, sharding, and legacy. If only Eq. (16) and a wall excess pass, report boundary-localized controller work without claiming that activity causes the CFT or tangent closing. The result is supplemental unless it quantitatively explains a Fig. 5 trend. Legacy parameter guidance is available in `topological_frustration_diagnostics/results/campaign_20260813_121340_Nx20_perfect_Ny20_22_24/`; its $N_x = 20$, $N_y = 20, 22, 24$ perfect-correction activity and spectral runs use only 25 trajectories, raster order, and short size-dependent times, so they are not production evidence. The corresponding formal reference is `topological_frustration_diagnostics/docs/numerical_diagnostics.tex`. Local click words and regularized-Choi endpoint spectra are not part of B2.

IX. TOPOLOGY AND WALL CONSTRUCTION

A. Campaign P1: quantitative revalidation of the BPJ bulk baseline

a. Claim and status. Status PLANNED. The fresh-ancilla $n_{\text{shell}} = 1$ realization quantitatively reproduces the postselection-free Chern-one bulk phase inherited from BPJ. This licenses the platform used by the wall campaigns; it is not the new universal result.

b. Dependencies and run matrix. Depends on V0. Use uniform periodic $L \times L$ systems with $L = 12, 16, 20, 24, 28, 32$, topological and trivial controller parameters, pure product-state initialization, and $S = 100$ per point. At $L = 20, 28$ also run max-mix initialization and repeat the pure-state topological and trivial points at $n_{\text{shell}} = 2$. This is the 20-family P1 block in the minimum execution matrix. Additional max-mix sizes are launched only if these two sizes or the wall purification campaign expose a bulk finite-size ambiguity.

c. Initialization, burn-in, and observation. Pure-state runs use the burn-in algorithm of Appendix C. The two max-mix sizes begin at $t = 0$ and have no burn-in; topology is evaluated only after the purity gap opens and observables become stationary.

d. Raw saves and estimator. Save purity spectra, local Chern markers, real-space Chern estimators, Bott data, density, and covariance checkpoints. For mixed covariances define $P_\xi = \Theta(C_\xi - \frac{1}{2})$ only if

$$\Delta_{\text{pur}, \xi} = \min_j |\nu_{j, \xi} - \frac{1}{2}| \quad (18)$$

exceeds the declared numerical and finite-size threshold. Compute each invariant per trajectory and then average its reported scalar/distribution.

e. Controls, statistics, and gate. Bootstrap trajectories. Compare at least two finite-size correction forms. Require the topological bulk marker to extrapolate to one, the trivial marker to zero, an open purity gap, and Bott index +1 in at least 95 of 100 accepted large-size trajectories. Require stability under the $n_{\text{shell}} = 2$ check.

f. Downgrade, figure, sharding, and legacy. If $n_{\text{shell}} = 1$ approaches a noninteger plateau, report that plateau or change the primary cutoff; do not claim convergence to one. Results map to Fig. 1(c,d). Shard by trajectory. A path-audited visual reference is `colab.lyapunov/figs/fig_chern_vs_cycle.png`; it is reference only.

B. Campaign W1: wall realization and transverse-convergence gate

a. Claim and status. Status PLANNED. An explicit topological–trivial controller interface and a support-terminated controller realize the same long-distance wall theory after transverse convergence.

b. Dependencies and run matrix. Depends on V1, B0, and P1; B1 is a nonblocking mechanism analysis. At $N_y = 48, 80$, use $N_x = 20, 24, 28$, $S = 100$, and compare: an explicit topological–trivial controller interface with both sides active and no support mask; a support-terminated `dw_truncation` controller with the exterior prepared and excluded according to the active-slab convention; a *separate* geometry- and update-count-matched trivial controller for each construction. The exact coupled spatial-mass Hamiltonian and P1 uniform topological runs are common benchmarks, not extra stochastic W1 protocols. Six of these 24 points coincide with accepted-width master families, leaving 18 new families. If $N_x = 20$ fails, extend the same grid upward in increments of four rather than preregistering unused widths.

c. Initialization, burn-in, and observation. Use independent product-state starts and independently determined burn-ins. Observe for a subsequent $2N_y$ window with checkpoints every $N_y/4$.

d. Raw saves and estimator. Save bulk-marker profiles, one- and two-wall entropy curves, charge FCS, correlators, tangent wall weights, and opposite-wall overlap. Compare asymptotic coefficients, not pointwise covariances.

e. Controls, statistics, and gate. The candidate $N_x = 20$ passes only if all headline coefficients agree with the next two widths within joint 95% intervals at both tested N_y , the fitted transverse localization length is well below wall separation, and opposite-wall low-mode weight is below 5%. The exact benchmark must also satisfy the hybridization inequality in Appendix C. Passing at $N_y = 48$ alone does not license a fixed- N_x thermodynamic claim because the required separation can grow as $\xi_{\perp} \log N_y$.

f. Downgrade, figure, sharding, and legacy. If $N_x = 20$ fails, all later campaigns use the smallest passing width. If no fixed width passes, use a joint N_x, N_y scaling. If the explicit-interface and support-terminated walls disagree, present them as distinct programmable walls rather than one universality class. Results map to Fig. 1(b,e) and Fig. S2. No compatible production legacy comparison is identified.

X. UNIVERSAL PREPARED-STATE DATA

A. Campaign S1: state CFT and charge sector

a. Claim and status. Status PLANNED. Stationary Born trajectories carry wall-localized complex-fermion state data with $c_{\text{edge}} = c_L + c_R = 1$ and a level-one $U(1)$ current algebra. This campaign does not establish handedness; chirality is licensed only after H1–H3.

b. Dependencies and run matrix. Depends on W1. Use the accepted N_x , Eq. (4), and $S = 100$. The primary wall and its matched control use both an onsite pure product state and max-mix initialization, the latter entering only after purification. At $N_y = 32, 48, 64$ also run the secondary wall construction and its separate matched control from pure product states. These are respectively the 12, 12, and 6 master-family blocks in the minimum matrix.

c. Initialization, burn-in, and observation. Pure runs use independent burn-ins and a separate $2N_y$ observation window. Save three stationary snapshots separated by at least $N_y/2$. Max-mix runs begin at $t = 0$ and enter this analysis only after the P2 stationarity gate.

d. Raw saves and estimator. For

$$X_L(\ell) = \log \left[\frac{L}{\pi} \sin \frac{\pi \ell}{L} \right], \quad (19)$$

fit the two-wall full-width interval to

$$\mathbb{E}[S_q(\ell)] = \frac{c_{\text{edge}}}{6} \left(1 + \frac{1}{q} \right) X_L(\ell) + b_q + \cdots, \quad q = 1, 2, 3, \quad (20)$$

using the full trajectory-bootstrap covariance across ℓ . Because one realization prepares one conditioned state, define $c_{\xi,q}$ trajectory by trajectory using the same preregistered fit window, design matrix, and linear estimator for every sample. Two co-primary summaries are reported: the coefficient c_q^{Born} fitted from the Born-averaged conditional entropy curve $\mathbb{E}_\xi[S_{\xi,q}(\ell)]$, and the full Born distribution of $c_{\xi,q}$ with its mean and whole-trajectory bootstrap interval. With a shared linear estimator, $c_q^{\text{Born}} = \mathbb{E}_\xi[c_{\xi,q}]$ algebraically, while displaying both the mean curve and the individual distribution separately tests ensemble scaling and trajectory typicality. Resolve the latter by the definite trajectory charge $N_{\mathcal{D},\xi}$ and report sector-conditioned means $c_q(N)$ wherever the sector count is sufficient. A broad $P(N)$ together with mutually compatible $c_q(N)$ is the direct test of a charge-fluctuating ensemble of critical states with shared infrared data. Do not average results from trajectory-dependent fit windows or model selections, and never extract a central charge from the entropy of the averaged covariance. Average multiple stationary snapshots within a trajectory before fitting or bootstrapping so they do not become independent samples.

e. Legacy-compatible entropy processing and production upgrade. The established repository estimator uses full-transverse strips

$$A(y_0, A_y) = [0, N_x) \times \{y_0, y_0 + 1, \dots, y_0 + A_y - 1\} \quad (21)$$

with periodic y , computes the entropy from the eigenvalues of the restricted one-body correlation matrix, and averages over all strip origins within each trajectory,

$$\bar{S}_{q,\xi}(A_y, t) = \frac{1}{N_y} \sum_{y_0=0}^{N_y-1} S_{q,\xi}[A(y_0, A_y), t]. \quad (22)$$

The legacy full-strip size campaign then averaged Eq. (22) over samples before an ordinary least-squares fit with an intercept against $X_{N_y}(A_y)$ on $A_y = 8, \dots, \lfloor N_y/2 \rfloor$ at a fixed final cycle. Its target von Neumann slope was $m_1 = 1/3$. Earlier wall-window contour scripts similarly summed the unnormalized contour over two-cell windows centered on each wall, averaged those curves over samples, and fitted the two complementary A_y branches separately.

Production retains this geometry, origin average, legacy fit window, intercept, and full-width target as the primary compatibility analysis, but preserves the sample axis before fitting. This follows the newer streaming repository path whose saved entropy array has axes (**sample**, **cycle**, **Ay**) after the y_0 average. For each q and trajectory, first average Eq. (22) over the declared stationary snapshots, then perform the common linear fit

$$\bar{S}_{q,\xi}(A_y) = m_{q,\xi} X_{N_y}(A_y) + b_{q,\xi}, \quad c_{\xi,q} = \frac{6m_{q,\xi}}{1 + 1/q}. \quad (23)$$

Thus $c_{\xi,1} = 3m_{1,\xi}$. Save $m_{q,\xi}$, $b_{q,\xi}$, $R_{q,\xi}^2$, residual vectors, and the number of fit points, along with the full per-trajectory entropy curves. The regression standard error across A_y used in legacy tables diagnoses deviation from the fitted shape; it is not the sampling uncertainty of the Born ensemble. All quoted uncertainties on c_q^{Born} , sector-conditioned $c_q(N)$, or distributional statistics come from resampling whole trajectories. Periodic origins and stationary snapshots remain correlated averages internal to one trajectory.

Repeat the fits after moving the lower endpoint from $A_y = 8$ by the B0-declared sensitivity increments, deleting the largest included A_y , and including the preregistered finite-size correction model. Check $S_{q,\xi}(A_y) = S_{q,\xi}(N_y - A_y)$ within tolerance for pure snapshots; fit the two branches separately only as a diagnostic. The primary wall-window result uses per-trajectory unnormalized contour sums before Born averaging, upgrading the older sample-averaged contour procedure without changing its spatial windows. The audited implementation references are `colab_large_entanglement_scaling_N20/src/strip_entropy_streaming_gpu.py`, `colab_charge_fluctuations/src/streaming_covariance_observables_gpu.py`, and `colab_charge_fluctuations/src/fit_streaming_covariance_scaling.py`; their legacy outputs remain comparison data rather than production inputs.

f. Production boundary-bulk correlator campaign. All correlator observables and fits are recalculated from the newly generated B0 covariances and S1 trajectory snapshots. Prior repository results motivate the observable and expose likely fit-window sensitivity, but none of their curves, averages, or fitted exponents enter the production estimates. This recalculation requires no additional Born families because it is performed on every covariance snapshot already generated for S1. Use the fixed-slice squared Green function

$$C_{G,\xi}(x, r_y) = \frac{1}{2N_y} \sum_{y,\mu,\nu} |C_{\xi;(x,y,\mu),(x,y+r_y,\nu)}|^2 \quad (24)$$

for every transverse slice x and every independent separation $1 \leq r_y \leq \lfloor N_y/2 \rfloor$. The exact B0 calculation and the stochastic S1 calculation use identical normalization, slice definitions, chord coordinate, candidate models, and fit windows.

For every trajectory retain the full $C_{G,\xi}(x, r_y)$ array and derive, without discarding the sample axis, the two individual wall curves, exterior-neighbor curves, distance-from-wall profiles, declared topological- and trivial-bulk averages, fixed wall-window sums, and the full- x average. Save all per-trajectory curves as well as the Born means, covariance across r_y , fit parameters, residuals, R^2 , point counts, and model-comparison scores. Source-position and stationary-snapshot averages remain correlated averages internal to a trajectory.

B0 fixes the primary distance window before the stochastic data are examined, using $r_y \in [3, 15]$ where the circumference permits. Fit the wall data to chord-power, exponential, chord-power-plus-floor, and exponential-plus-floor models; apply the same models to matched bulk data rather than assuming exponential decay. Endpoint and window sweeps are sensitivity diagnostics and may not select the reported exponent. Report the Born-averaged fit and the complete trajectory-level exponent distribution, including its dependence on the trajectory charge sector, exactly as for the central-charge analysis. The primary conclusion is spatial: stable algebraic behavior should be wall localized and absent from the bulk. Quote a precision exponent near two only if it is stable under the preregistered window and model checks; otherwise retain the freshly calculated boundary-bulk contrast as qualitative support and let entropy and charge fluctuations carry the universal claim. Bootstrap whole trajectories.

The audited references `exact_DW_benchmark_notes/exact_DW_GS_benchmark_revtext.tex` and `colab_charge_fluctuations/analysis_outputs/correlator_scaling_diagnostics/` are used only to document what was tried before and to preregister failure modes. Every plotted curve and quoted number in the new manuscript is regenerated under the production protocol.

Test whether the contour integrated over a declared single-wall window carries half that coefficient. This window sum is a diagnostic decomposition of the full-strip entropy, not the entropy of an independent wall subsystem. Save every trajectory's subsystem spectra, entropy curves, unnormalized spatial contour $s_{A,\xi}(x, y)$, charge-generating function, wall-projected connected density structure factor, and globally normalized real-space correlator. Form $s_A(x, y) = S^{-1} \sum_{\xi} s_{A,\xi}(x, y)$ without rescaling individual contours, and verify $\sum_{x,y} \bar{s}_A = \mathbb{E}[S_A]$. Average S_q , quantum cumulants, structure factors, and squared correlations only after their per-trajectory evaluation. The interval variance and its small- k structure factor are primary current diagnostics; the real-space squared Green function is an appendix cross-check. Appendix A fixes the FCS conventions.

Separate quantum charge fluctuations from classical record-to-record profile fluctuations. For every interval A save

$$F_{\xi}^q(A) = \langle N_A^2 \rangle_{\xi} - \langle N_A \rangle_{\xi}^2 = \text{Tr}[C_{\xi,A}(1 - C_{\xi,A})], \quad (25)$$

$$F^{\text{cl}}(A) = \text{Var}_{\xi}(\langle N_A \rangle_{\xi}), \quad (26)$$

and verify the law of total variance

$$\text{Var}(N_A) = \mathbb{E}_{\xi}[F_{\xi}^q(A)] + F^{\text{cl}}(A). \quad (27)$$

Only the logarithmic coefficient of $\mathbb{E}_{\xi}[F_{\xi}^q(A)]$ is used to infer the $U(1)$ current level. The classical term is not purely sector wandering. Let $\mathcal{D}$ denote the declared dynamical subsystem—the active slab for support termination and the full system for the explicit interface—and let $N_{\mathcal{D}}$ be its charge. For a definite-charge input, each pure conditioned trajectory has a definite $N_{\mathcal{D}}$ and

$$F^{\text{cl}}(A) = \mathbb{E}_{N_{\mathcal{D}}}[\text{Var}_{\xi|N_{\mathcal{D}}}(\langle N_A \rangle_{\xi})] + \text{Var}_{N_{\mathcal{D}}}(\mathbb{E}_{\xi|N_{\mathcal{D}}}[\langle N_A \rangle_{\xi}]). \quad (28)$$

The first term is fixed-sector shape variation and the second is the contribution from sector wandering. Max-mix trajectories enter this decomposition only after both the global purity deficit and the within-trajectory quantum variance of $N_{\mathcal{D}}$ fall below declared tolerances.

To test where the total-sector fluctuations couple spatially, define

$$q_{\xi x} = \sum_{(y,\mu):(x,y,\mu) \in \mathcal{D}} \langle n_{xy\mu} \rangle_{\xi}, \quad N_{\mathcal{D},\xi} = \sum_x q_{\xi x}. \quad (29)$$

Save the complete transverse covariance matrix and its signed additive row sum,

$$K_{xx'}^{\text{cl}} = \text{Cov}_{\xi}(q_{\xi x}, q_{\xi x'}), \quad \kappa_{\text{cl}}(x) = \sum_{x'} K_{xx'}^{\text{cl}} = \text{Cov}_{\xi}(q_{\xi x}, N_{\mathcal{D},\xi}), \quad \sum_x \kappa_{\text{cl}}(x) = \text{Var}_{\xi}(N_{\mathcal{D},\xi}). \quad (30)$$

Also save the nonnegative onsite-heterogeneity profile $v_{\text{cl}}(x) = \sum_{y,\mu} \text{Var}_{\xi}(\langle n_{xy\mu} \rangle_{\xi})$. Report signed κ_{cl} without taking its absolute value, its signed partial sums in fixed wall windows, and its absolute leakage outside those windows. Full-system charge bookkeeping is saved separately. The profile v_{cl} omits inter-site covariances, so it neither sums to $\text{Var}(N_{\mathcal{D}})$ nor defines a variance weight; use it only as a descriptive localization control. Match every system-charge change to the signed gain/loss record. These quantities test whether sector wandering is coupled to the walls; they are not themselves universal CFT coefficients.

g. Matched entropy–variance spectral test. For every trajectory, stationary snapshot, origin, and interval used in the entropy fit, compute the quantum internal variance from the same restricted covariance and perform the same internal snapshot and origin averages. With

$$\bar{F}_\xi^q(A_y) = f_\xi X_{N_y}(A_y) + b_{F,\xi} + \dots, \quad (31)$$

define the paired trajectory statistic and its Born mean by

$$\delta_{SF,\xi} = m_{1,\xi} - \frac{\pi^2}{3} f_\xi, \quad \delta_{SF}^{\text{Born}} = \mathbb{E}_\xi[\delta_{SF,\xi}] = a_S - \frac{\pi^2}{3} a_F. \quad (32)$$

The primary analysis fits the paired residual curve

$$R_{SF}(A_y) = \mathbb{E}_\xi \left[\bar{S}_{1,\xi}(A_y) - \frac{\pi^2}{3} \bar{F}_\xi^q(A_y) \right] = \delta_{SF}^{\text{Born}} X_{N_y}(A_y) + b_{SF} + \dots \quad (33)$$

with the full paired covariance across A_y and a whole-trajectory bootstrap. Report a_S/a_F only as a secondary statistic because a ratio is unstable when its denominator is uncertain. For Rényi index q , also test the free-fermion prediction

$$a_{S_q} - \frac{\pi^2}{6} \left(1 + \frac{1}{q} \right) a_F = 0. \quad (34)$$

Appendix B derives these relations and their limits. This is an observer analysis of the existing S1 master covariances and launches no new trajectory family.

h. Controls, statistics, and gate. Use a trajectory bootstrap with the full ℓ covariance matrix. Predeclare fit windows from B0 and repeat fits after deleting each endpoint window and the smallest size. Require $c_{\text{edge}} = 1$ for all three Rényi indices, the factor-of-two relation between one and two walls, and level $k = 1$ from both interval variance and the small- k structure factor. Using the same fit windows and finite-size model, require the thermodynamic extrapolation of $\delta_{SF}^{\text{Born}}$ to be compatible with zero at 95% confidence and stable under the declared endpoint and smallest-size deletions. Apply the same paired test to the exact-Hamiltonian benchmark and require no false logarithmic signal in the matched trivial control. The wall squared-Green-function exponent must be compatible with two on the benchmark-selected window and remain stable under the declared sensitivity checks; no matched bulk slice may support a comparably stable algebraic fit. If the exponent remains window sensitive, downgrade it to a boundary-localization diagnostic rather than tuning the window. Require a log-chord model to outperform an area-law constant in the wall geometry and not in the trivial control.

i. Downgrade, figure, sharding, and legacy. If only entropy is compatible with one, say “consistent with a $c = 1$ wall” and do not claim a charge-resolved CFT. If c and k are individually stable but Eq. (32) fails, report them separately and do not identify the wall as a minimal charged $U(1)_1$ sector. This campaign supplies all of Fig. 2. Shard by trajectory; source-position averages are internal correlated averages. A path-audited visual reference is `colab_large_entanglement_scaling_N20/docs/figures/slope_and_one_minus_r2_vs_Ny.png`. The contour is a localization and sum-rule check, not a standalone novelty claim; BPJ already used entanglement contours to identify slow domain-wall structure. No compatible legacy dataset contains both the full-strip entropy and the matched within-trajectory interval variance: the saved streaming campaign contains the former, while the saved max-mix total variance is a purification observable rather than the S1 current-algebra variance.

B. Campaign S2: topological-to-trivial sweep

a. Claim and status. Status PLANNED. The logarithmic wall sector and near-marginal tangent sector occur when the two sides have different bulk topology.

b. Dependencies and run matrix. Calibrate α_c from the exact uniform Hamiltonian. Fix $\alpha_{\text{out}} = 30$ and take

$$\alpha_{\text{in}} = \alpha_c + \{-1, -0.5, -0.25, -0.125, 0.125, 0.25, 0.5, 1\}, \quad (35)$$

plus $\alpha_{\text{in}} = 1$, at $N_y = 24, 32, 48, 64$, the accepted N_x , and $S = 100$.

c. Initialization, saves, estimator, and controls. Use independent pure-state burn-ins. Save the bulk marker and purity gap, entropy and charge coefficients, correlator, and tangent gap. Do not fit the bulk-critical point itself to a pure wall CFT. Compare deep topological, crossover, and deep trivial regimes.

d. Statistics and acceptance gate. Bootstrap the joint location of the topological-marker change, disappearance of the logarithmic coefficient, and opening of the tangent gap. The gate requires a common finite-size crossover interval, not identical finite-size pseudocritical points.

e. Downgrade, figure, sharding, and legacy. Preemptive wall crossover is allowed and must be reported. This is Fig. 2(e) if compact; otherwise Fig. S3. Shard by trajectory. No compatible legacy comparison is identified.

XI. TANGENT DYNAMICS AND PURIFICATION

A. Campaign T1: quenched tangent spectrum and localization

a. Claim and status. Status PLANNED. The differential of the normalized realized covariance map has a wall-localized near-marginal sector whose gap closes with N_y although exact projectors make a conventional square transfer product singular.

b. Dependencies and run matrix. Depends on W1 and the burn-in pilot. Use the accepted N_x , Eq. (4), $S = 100$, wall and matched trivial controls, and enough tangent vectors to retain at least 12 physical modes nearest zero. At $N_y = 32, 48$, double the tangent-frame rank and repeat at least four independently randomized initial tangent frames on the same replayed physical records.

c. Initialization, burn-in, and observation. Burn in the physical trajectory without accumulating exponents. At the accepted burn-in, initialize a common physical tangent subspace, propagate an alignment interval of N_y cycles whose QR logs are discarded, reset the log sums without resetting the physical trajectory or aligned frame, and accumulate initially for $16N_y$ cycles. Checkpoints occur at $N_y, 2N_y, 4N_y, 8N_y, 16N_y$. At each fixed N_y , extrapolate the tracked rates in $1/T$ (and at least one alternative correction) or extend until the final independent blocks agree within tolerance. No N_y scaling fit begins before this fixed-size temporal limit is passed.

d. Raw saves and estimator. Save per-cycle QR log increments, finite/null masks, gauge directions, aligned low vectors, orthogonality residuals, and stable cocycle snapshots. A snapshot consists of the physical covariance at its left endpoint, the ordered realized branch/channel word for the window, the aligned input and output QR frames, block- R factors with their diagonal log scales, and a replay hash. On validation sizes also save the explicit window product and its SVD; at production sizes reconstruct it from the stabilized factors rather than multiplying a long ill-conditioned dense word. For max-mix bridge snapshots save the reduced endpoint Choi frames A_K, B_K and Gram matrix needed to verify $\tilde{J}_K^\dagger \tilde{J}_K = 4C_K(1 - C_K)$. For trajectory s , define the one-leg half-Jacobian growth rates

$$\ell_{j,s}(T) = \frac{1}{T} \sum_{\tau=1}^T \log |(R_{\tau,s})_{jj}|, \quad \Delta_{1\text{leg}}(T) = \min_{j \in \text{physical}} |\bar{\ell}_j(T)|. \quad (36)$$

The primary quenched spectrum averages log stretches at continuously tracked Oseledec index before taking the gap. If decay-rate notation is desired, use $\gamma_j = -\ell_j$ and state the sign convention. The full covariance-tangent map is a congruence, so its exponents are pair sums $\ell_a + \ell_b$ (or decay rates $\gamma_a + \gamma_b$), not the one-leg spectrum itself. The mean and median of per-trajectory minimum gaps are reported as sensitivity diagnostics. Exact null and normalization directions are removed before defining the physical reference exponent.

For a near-degenerate low set $\mathcal{L}_s$ of rank r_s , localize its projector rather than a basis-dependent QR vector. The primary density is

$$\rho_s(x, y) = \frac{1}{r_s} \sum_{a \in \mathcal{L}_s, \mu} |v_{s,a,\mu}(x, y)|^2, \quad \sum_{x,y} \rho_s(x, y) = 1. \quad (37)$$

Never normalize the orbital components independently. Resolve both walls by diagonalizing a projected transverse-position operator within $\mathcal{L}_s$ or by reporting left/right-window weights; do not select whichever individual QR vector looks most localized.

e. Controls, statistics, and gate. Compare aN_y^{-z} , a free-intercept power law, and a gapped constant plus corrections, using at least five sizes and bootstrap fits. Require the wall gap to have zero intercept, preferably $z = 1$, while the matched trivial control retains a positive intercept. Require the fixed- N_y temporal extrapolation above, initial-frame and rank convergence, near-degenerate-subspace interface weight above 70%, representation of both walls, and transverse localization satisfying W1.

f. Downgrade, figure, sharding, and legacy. If the intercept is unresolved, call the result finite-size softening rather than a thermodynamic closing. Results supply Fig. 5(b,c). Shard whole trajectories; never split one QR history across unordered shards. References are `colab.lyapunov/figs/final_gap-vs-Ny.lyapunov-pc.png` and `colab.lyapunov/figs/fig_eigvec_heatmap-pc-avg.pdf` only. Their observation time scaled with N_y , so they motivate parameters and localization windows but cannot by themselves separate finite-time drift from a thermodynamic gap closing.

B. Campaign P2: max-mix purification and recordwise tangent bridge

- a. Claim and status.* Status PLANNED. The tangent singular spectrum quantitatively controls slow purification of a maximally mixed state at the wall.
- b. Dependencies and run matrix.* Depends on T1. Use the accepted N_x , Eq. (4), $S = 100$, wall and matched trivial controls, and a duration of $16N_y$ cycles.
- c. Initialization, burn-in, and observation.* Set $C(0) = 1/2$. There is *no burn-in*: purification begins at $t = 0$. Save every early cycle and logarithmically spaced late times. Any stationary-window analysis is a separate post-purification operation.
- d. Raw saves and estimator.* Save global entropy, $\text{Tr}[C(1 - C)]$, purity gap, spatial entropy contour, Chern marker, charge cumulants, and matched tangent singular data. For each trajectory propagate the *same* max-mix record and the full start-to- t cocycle, and validate

$$\tilde{J}_t^\dagger \tilde{J}_t = 4C_t(1 - C_t) \quad (38)$$

record by record at the individual-mode level and again after conversion to entropy. This identity does not directly identify the stationary pure-start T1 spectrum; compare the two only after their record, initialization, and time-window differences are controlled.

Use the entropy contour and the matched wall-off run to define a wall-excess residual $S_{\text{wall}}^{\text{res}}(t, L)$. Fit

$$S_{\text{wall}}^{\text{res}}(t, L) = L^y \mathcal{F}(t/L^z), \quad (39)$$

testing the per-length hypothesis $y = 1$ and a free y . A global entropy initialized from max-mix is not collapsed without this amplitude factor or an explicitly justified normalization. Fit z independently and compare it with the appropriate pair-sum tangent prediction; do not force agreement with the one-leg T1 gap.

e. Controls, statistics, and gate. Require per-record algebraic agreement within numerical tolerance, compatible tangent and purification scaling after the one-leg/pair-sum conversion, bulk purification faster than the wall sector, and a collapse stable to time-window and amplitude-normalization changes.

f. Downgrade, figure, sharding, and legacy. If the exact bridge passes but no universal collapse does, retain the identity and report the measured time dependence without a universal exponent. Results supply Fig. 5(d,e). Shard by complete trajectory. No production-compatible legacy comparison is required.

XII. EXACTLY THREE CHIRALITY ROUTES

The three routes use different clocks and prove different statements. Modular time probes the prepared state's entanglement generator; monitored circuit time probes physical Born-ensemble response; twist angle defines an auxiliary integer spectral-flow index. No one route substitutes for another, and none is called a quantized transport coefficient.

A. Campaign H1: modular charge drift

- a. Claim and status.* Status PLANNED. The wall-reduced modular generator is handed, with opposite signs on oppositely oriented walls.
- b. Dependencies and run matrix.* Depends on S1. Use $N_y = 32, 48, 64$, the accepted N_x , both oppositely oriented walls, the trivial control, and $S = 100$. A separately reversed-Chern controller is an escalation only if the two-wall orientation test is ambiguous; it is not a baseline trajectory family.
- c. Initialization, burn-in, and observation.* Use independent stationary snapshots from S1. Define the wall subsystem before examining the result and construct

$$h_{\xi, w} = \log \frac{1 - C_{\xi, w}}{C_{\xi, w}}. \quad (40)$$

Launch identical localized packets and stop the modular-time fit before wraparound or loss of wall support.

d. Raw saves and estimator. Save modular spectra, clipping diagnostics, signed packet density, wall retention, width, periodic center of mass, and per-trajectory fitted velocity. Average signed densities and velocities only after per-trajectory construction.

e. Controls, statistics, and gate. Require opposite velocity signs with nonoverlapping 95% intervals, sign reversal with wall orientation, sustained wall localization, and no stable trivial-control drift. Vary the wall window, modular spectral cutoff, and fit interval.

f. Downgrade, figure, sharding, and legacy. This proves modular handedness, not monitored-time transport. It supplies Fig. 3(a). Reference-only material exists under `notebooks/modular_charge_spreading/`.

B. Campaign H2: physical Born response $\chi(k, \omega)$

a. Claim and status. Status PLANNED. The complete Born ensemble has a wall-localized causal response with a single low-energy velocity sign on each wall.

b. Dependencies and run matrix. Depends on W1 and V0 response validation. Use $N_y = 32, 48, 64$, $S = 100$, eight source positions and four stationary time origins per trajectory, and response horizon $N_y/2$. Random order is primary; raster and reverse raster are run at $N_y = 48$.

c. Initialization, burn-in, and observation. Each of the 100 trajectories receives an independent burn-in. Apply an infinitesimal local density-potential kick only after stationarity. Sources and time origins inside one trajectory are correlated variance-reduction samples, not new trajectories.

d. Raw saves and estimator. For an observable O , evaluate the exact decomposition

$$\partial_\epsilon \mathbb{E}_{\xi, \epsilon}[O_{\xi, \epsilon}]|_0 = \mathbb{E}_\xi[\partial_\epsilon O_\xi] + \mathbb{E}_\xi[O_\xi \partial_\epsilon \log p_\xi]. \quad (41)$$

Save the conditional derivative, score term, total signed response, and cumulative score separately. Validate Eq. (41) against brute-force $\pm\epsilon$ ensembles on small systems. Compute

$$\chi_w(k, \omega) = \sum_{t, y} e^{i\omega t - iky} w_t \chi_w(y, t), \quad (42)$$

with a recorded window w_t . Zero padding is visual only; ridge fits use native resolution. Define in advance a low-energy wedge asymmetry between $\omega k > 0$ and $\omega k < 0$.

e. Controls, statistics, and gate. Bootstrap whole trajectories. Require a low-energy linear ridge, opposite slopes on the two walls, a stable sign under random and reversed schedules, subdominant wrong-chirality weight, wall localization, and no corresponding trivial-control ridge. Both terms in Eq. (41) must be included in the headline panel.

f. Downgrade, figure, sharding, and legacy. If only the fixed-record term is chiral, label it conditional response and do not claim physical Born-ensemble chirality. Results supply Fig. 3(b), with the two wall orientations shown inside one compound panel. Shard trajectories while keeping all sources together. No compatible legacy $\chi(k, \omega)$ result including the score term is identified.

C. Campaign H3: frozen-record integer twist flow

a. Claim and status. Status PLANNED. A continuous flux family constructed from a single realized record has unit wall-resolved spectral flow.

b. Dependencies and run matrix. Depends on S1 and V0 twist validation. Use $N_y = 24, 40, 64$, $S = 100$, and

$$\phi_m = 2\pi m/64, \quad m = 0, \dots, 64. \quad (43)$$

At one representative size repeat 33, 65, and 129 points and move the twist seam.

c. Initialization, burn-in, and observation. Sample each record physically at $\phi = 0$, including an independent burn-in. Freeze that entire discrete record. Insert the $U(1)$ phase into every controller mode crossing the periodic y seam before support truncation, and replay the same record at every ϕ . Never draw an independent Hamiltonian or record at each twist.

d. Raw saves and estimator. Save bit-packed records, $p_\xi(\phi)$ in log form, purity gaps, covariances, wall entanglement eigenvalues and vectors, overlap matrices, tracking assignments, crossing tables, and 2π closure residuals. Track

$$\varepsilon_a(\phi) = \log \frac{1 - \nu_a(\phi)}{\nu_a(\phi)} \quad (44)$$

by maximum-overlap subspace transport and count signed zero crossings separately at the two walls.

e. Controls, statistics, and gate. Require nonzero branch weight for the full circle, closure up to the large gauge transformation, $|\nu_{\text{flow}}| = 1$ in at least 95 of 100 records, opposite signs on the two walls, zero flow in the trivial control, and stability to grid, seam, tracking tolerance, and reference level inside the entanglement gap.

f. Downgrade, figure, sharding, and legacy. A record whose branch weight becomes exactly zero does not define this flux family and may not be replaced by another record. If too many fail, report the bulk Bott number and nonquantized spectral behavior only. Results supply Fig. 3(c). Shard by frozen record, keeping its complete ϕ circle on one shard. The directory `tangent_cocycle_flux_snapshot/` is code reference only; no compatible saved production figure is identified.

XIII. TRAJECTORY-RESOLVED UNIVERSALITY

A. Campaign R1: Born self-information and charge-resolved record spectrum

a. Claim and status. Status PLANNED. After an explicit replica/transfer normalization is validated, independent universal data $c_{\text{eff}}^{\text{rec}}$ and k_{rec} can be reconstructed from the classical measurement record without trajectory-state tomography and then compared with the state data.

b. Dependencies and run matrix. Depends on W1, T1, and an independently calibrated record velocity. Use the accepted $N_x, N_y = 16, 20, 24, 28, 32, 40, 48, 64, S = 100$, a matched trivial controller with identical active-mode count, and a post-burn observation of at least $32N_y$ cycles. Analyze temporal aspect ratios $T/N_y \in \{2, 4, 8, 16, 32\}$ from nested blocks without treating them as independent trajectories.

c. Initialization, burn-in, and observation. Every trajectory is independently burned in. Divide the observation into declared nonoverlapping blocks, retain the trajectory as the bootstrap unit, and do not include initialization in the stationary estimator. Estimate stationarity before combining block origins.

d. Raw saves and estimator. For each record save its self-information

$$I_\xi(t_1, t_2) = - \sum_{j \in [t_1, t_2]} \log p(m_j | \xi_{<j}). \quad (45)$$

Its Born mean is a Shannon self-information, not automatically a CFT free energy. The required bridge is the replica partition function

$$Z_n(L, T) = \sum_{\xi} p_{\xi}^n, \quad -\partial_n \log Z_n|_{n=1} = \mathbb{E}_{\xi}[I_{\xi}]. \quad (46)$$

Derive the transfer normalization and boundary conditions of Z_n for the implemented schedule before converting a coefficient to a central charge. Form a wall-minus-matched control subtraction with identical event count and define f_{def} per wall, per monitored cycle, and per wall length. First fit the agnostic form

$$f_{\text{def}}(L, T/L) = f_{\infty}(T/L) - \frac{A_{\text{rec}}(T/L)}{L^2} + \frac{b_4(T/L)}{L^4} + \dots \quad (47)$$

Only if the replica derivation fixes the chiral-cylinder convention may one identify the large-aspect-ratio amplitude as

$$A_{\text{rec}} = \frac{\pi v_{\text{rec}} c_{\text{eff}}^{\text{rec}}}{12} \quad (48)$$

per wall. For both walls before division by wall number, the coefficient doubles. The estimator is the arithmetic Born mean of $-\log p_{\xi}$, never the logarithm of an average probability.

Calibrate v_{rec} in two ways: spacetime collapse of connected local record correlations and momentum/twist dispersion of a charge-identified low transfer sector. Fit the record current level independently through

$$h_{q,n}^{\text{rec}} = \frac{q^2}{2k_{\text{rec}}} + n, \quad q \in \mathbb{Z}, \quad n \in \mathbb{N}_0, \quad (49)$$

including descendant degeneracies. At minimum resolve the $q = \pm 1$ primary and a neutral current or descendant sector. Test $k_{\text{rec}} = 1$ and $c_{\text{eff}}^{\text{rec}} = c_{\text{edge}}$ only after the independent record fits.

e. Controls, statistics, and gate. Use late-time block increments and trajectory bootstraps. Require stationarity, stable L^{-2} coefficients after deleting small sizes and adding L^{-4} , convergence in temporal aspect ratio, agreement of the two velocity calibrations, a vanishing matched-trivial defect coefficient, and at least two nontrivial charge-resolved dimensions consistent with one common k_{rec} . The normalization must also reproduce an exactly enumerated small-system Z_n and its derivative at $n = 1$.

f. Downgrade, figure, sharding, and legacy. Without the replica/transfer derivation report A_{rec} as a defect self-information coefficient, not a central charge. With the derivation but without an independent velocity report $v_{\text{rec}}c_{\text{eff}}^{\text{rec}}$, not $c_{\text{eff}}^{\text{rec}}$. One low gap is not a CFT spectrum. Results supply Fig. 4. Shard by whole trajectory. No previous cumulative/no-burn central-charge output is compatible with this estimator; legacy comparison is therefore none.

XIV. MEAN, POSTSELECTION, AND NOISE CONTROLS

A. Campaign M1: trajectory ensemble versus mean channel

a. Claim and status. Status PLANNED. This campaign determines which wall signatures survive without access to the measurement record and tests the sharper possibility that a gapped unconditional map coexists with a closing conditioned tangent sector. It does not assume in advance that the mean channel is trivial or noncritical.

b. Three mean evolutions that must not be conflated. Let $P_j = |\chi_j\rangle\langle\chi_j|$, $Q_j = \mathbf{1} - P_j$, target s_j , and $\eta_j = 2s_j - 1$. For the fresh-ancilla perfect-correction primitive, averaging only the Born outcome gives the exact one-step replacement channel

$$\mathcal{E}_j(\rho) = |s_j\rangle\langle s_j|_{\chi_j} \hat{\otimes} \text{Tr}_{\chi_j} \rho, \quad \mathcal{R}_j(G) = Q_j G Q_j + \eta_j P_j. \quad (50)$$

Equation (50) uses the graded fermionic tensor factorization and is applied to the parity-even, number-conserving states of the present protocol. For a prescribed chronological schedule $\pi = (j_1, \dots, j_M)$, the exact discrete mean sweep is

$$\mathcal{R}_\pi = \mathcal{R}_{j_M} \circ \dots \circ \mathcal{R}_{j_1}. \quad (51)$$

It is the Born average of the implemented perfect-correction circuit at fixed external schedule. Since each replacement maps a Gaussian state to a Gaussian state, $\bar{\rho}_\pi$ remains Gaussian. An additional mixture over randomized schedules need not be Gaussian, although its covariance remains the corresponding average of Eq. (51). A history with a newly randomized word each cycle is a nonautonomous sequence of affine maps; it does not possess one stationary single-sweep multiplier spectrum. Spectral multipliers are defined only for a periodically repeated word, an explicitly averaged/symmetrized autonomous map, or as exponents of a separate mean-map cocycle.

Define separately the Poissonized continuous-time model

$$\mathcal{L}_{\text{pc}} = \sum_j r_j (\mathcal{E}_j - \mathbf{1}), \quad \dot{G} = \sum_j r_j (Q_j G Q_j + \eta_j P_j - G) \equiv \mathcal{A}[G] + B. \quad (52)$$

This is a valid Lindblad generator for independent Poisson applications of the local replacement channels. Use $r_j = 1$ when one unit of continuous time is to contain one expected firing of each channel. It is not the logarithm of the ordered sweep, $\mathcal{L}_{\text{pc}} \neq \log \mathcal{R}_\pi$ in general, and agreement with the discrete circuit is an empirical coarse-graining test.

The bath-density generator in `CI.Lindblad.DW.py`, with one occupation $\bar{n}_a$ controlling both gain and loss and an optional decoherence term, is a third model. It describes the earlier Markovian-bath approximation, not the exact mean of the fresh target-occupation ancillas. Its affine-vectorization, partial Fourier transform, momentum-transfer sectors, and pulse-response routines are useful infrastructure, but production must rebuild P_j from the canonical controller and implement Eqs. (50) and (52). Do not tune $\bar{n}_a$ and call the result perfect correction.

c. Dependencies and run matrix. Depends on V0, V1, W1, S1, and T1. Validate the density-matrix channel by exact Fock-space enumeration on 2×2 and 2×3 systems and the covariance implementation on 4×6 and 6×8 . Production consumes every size already present in the primary and secondary wall/control master families; it does not fill missing protocol-size cells merely to make a rectangular M1 grid. For random serial order, use the same $S = 100$ external schedule seeds as the paired trajectory family and compute one deterministic fixed-schedule mean evolution per seed. The V1 raster and reverse-raster masters supply the ordered-word comparison at $N_y = 48$. If the one-record-per-schedule decomposition cannot distinguish Born from schedule variance, a separate fixed-schedule $S = 32$ pilot on the two smallest primary-wall sizes may be launched, but it is not part of the minimum campaign or any manuscript gate.

For spectral work, begin with $N_x = 8, 12$ and $N_y = 16, 24, 32, 48, 64$ and proceed to the accepted production width only if matrix-free residuals and memory estimates pass. Never construct the dense $N_{\text{sp}}^2 \times N_{\text{sp}}^2$ covariance superoperator at production sizes. Apply $\mathcal{A}$ through rank-one projector actions and use matrix-free Arnoldi/Krylov methods. Translation invariance along the wall permits exact momentum-transfer sectors $q = k_y - k'_y$ only after verifying $[\mathcal{R}_\pi, \mathcal{T}_y] = 0$. This holds for the translation-invariant Poisson sum with uniform rates and for an explicitly translation-symmetrized autonomous map, but a generic random-serial or raster word breaks one-site translation. Use real space or the actual supercell momentum in that case. When the symmetry is present, the affine source lies in $q = 0$.

d. Initialization, burn-in, and raw saves. Use exactly the same initial covariance, controller-frame hash, domain-wall definition, schedule word, and channel ordering as the paired trajectory run. The deterministic discrete map uses the trajectory burn-in window. The Poissonized model begins at the same state at $t = 0$ and is evolved long enough to satisfy a separate stationarity gate; it does not borrow the discrete burn-in in cycle units.

Save fixed-schedule $\bar{C}_\pi(t)$, the schedule-averaged covariance, residuals of the affine fixed-point equation, the homogeneous discrete sweep action

$$\delta G \mapsto Q_{j_M} \cdots Q_{j_1} \delta G Q_{j_1} \cdots Q_{j_M}, \quad (53)$$

and the leading eigenpairs of both this map and $\mathcal{A}(q)$ when the corresponding map is autonomous. For randomized histories save the complete map word and accumulate mean-map-cocycle exponents rather than diagonalizing a fictitious average word. Save Hermiticity, spectral-interval, trace/charge, complete-positivity small-system, Fourier-block reconstruction, and Krylov residual diagnostics.

e. Linear-observable and record-rate identities. Compare the deterministic mean covariance with $S^{-1} \sum_{\xi=1}^S C_\xi(t)$ for all one-body densities, currents, charges, and selected matrix elements. Before channel j , the wrong-outcome probability is

$$p_j^{\text{wrong}} = \frac{1 - \eta_j \text{Tr}(P_j G)}{2}. \quad (54)$$

Use this identity to predict the mean correction-event density and signed system-charge transfer, then compare with the trajectory-resolved record ensemble globally and as a transverse wall profile. Connected temporal correlations and record large deviations are not fixed by the one-time mean covariance and remain separate observables.

f. State and topology diagnostics. At fixed schedule compare three nonlinear levels:

$$\mathbb{E}_{\xi|\pi}[S(C_{\xi,A})], \quad S(\bar{C}_{\pi,A}), \quad S_G(\mathbb{E}_\pi[\bar{C}_{\pi,A}]). \quad (55)$$

The middle quantity is the exact entropy of the fixed-schedule Gaussian mean state. The last is only the entropy of the Gaussian state sharing the schedule-averaged covariance; on small Fock-space systems compare it with the actual entropy of the schedule mixture and save Wick-factorization residuals. Likewise report $\mathbb{E}_\xi[|C_\xi(r)|^2]$ and $|\mathbb{E}_\xi C_\xi(r)|^2$ separately.

Resolve the stationary mean-covariance occupation spectrum $\nu_a(k_y)$, wall weights of states nearest $\nu = 1/2$, and bulk and wall purity gaps. A projector obtained by flattening occupations above $1/2$ may be assigned a Chern marker only where a purity gap is open. If the wall branch crosses $1/2$, quote bulk-window markers and spectral flow rather than a globally gapped mixed-state invariant. A covariance Chern marker of a non-Gaussian schedule mixture is a one-body diagnostic, not automatically a density-matrix invariant.

g. Mean relaxation and response. For the discrete map report multipliers $\mu_a(q)$ and cycle exponents $-\log |\mu_a|$; for the Poissonized map report eigenvalues $\lambda_a(q)$ and

$$\Delta_{\text{mean}} = \min_{q, a \notin \text{stationary}} [-\Re \lambda_a(q)]. \quad (56)$$

Identify every excluded stationary or symmetry mode and localize each slow eigenoperator in x . The q label is used only after the translation test above; otherwise report real-space or supercell eigenmodes. In the support-terminated geometry construct and diagonalize the relaxation map on the same active slab used by the conditioned tangent calculation. The full-space map has frozen-exterior unit multipliers, and diagonalizing it first and discarding those modes afterward is not a physical gap definition. These spectra are not the conditioned tangent Lyapunov spectrum. The central comparison is whether $\Delta_{\text{mean}} = O(1)$ while the wall-localized conditional tangent gap closes.

Use the homogeneous maps to evolve a physical small perturbation δG around the stationary mean. Prefer a local unitary kick or a validated symmetric source that keeps $0 \leq C \leq 1$ over simply adding an occupation to a saturated mode. Measure the wall-projected $\chi_{\text{mean}}(k, \omega)$, drift on both wall orientations, DW-off response, and random/raster/reverse controls. A complex linear branch or directed response would show that unconditional one-body dynamics retains chirality; it would not by itself restore the trajectory entanglement CFT or define quantized transport.

h. Controls, statistics, and gate. Linear observables and Eq. (54) must agree with the Born sample within simultaneous trajectory confidence intervals. Require exact-map agreement with Fock-space enumeration, fixed-point residual below tolerance, $0 \leq \bar{C} \leq 1$, time-step convergence for any RK implementation, and agreement between full and q -sector spectra on small systems. Compare exact discrete, Poissonized, and legacy bath models side by side without fitting their clocks after viewing the largest-size result.

This campaign is a conceptual discriminator, not a gate on the existence of the trajectory CFT. Do not describe the averaged state with an effective-temperature interpretation unless one common β fits a finite-temperature CFT correlator, occupation spectrum, and a fluctuation-dissipation or detailed-balance test. If only exponential decay or spectral broadening is observed, use finite correlation length or dephasing scale.

i. Downgrade, figure, sharding, and legacy. If the mean map is gapped and noncritical, use one compact main-text panel contrasting its finite correlation length with the trajectory CFT and place its full spectrum in the Supplement. If it has a wall-localized closing or chiral response, promote the conditioned/unconditioned comparison because that changes the central interpretation. The legacy implementation is `CI_Lindblad_DW.py`; its existing occupation spectra, local-marker, current, entanglement-proxy, and pulse-response outputs guide parameter and window selection only. Its defaults use an untruncated or separately truncated spatial-mass construction and do not count as data for either canonical wall protocol until the controller frames and parameters match exactly.

B. Campaign M2: postselection and imperfect feedback

a. Claim and status. Status PLANNED. The new wall-CFT, chirality, tangent, and record signatures occur in the physical Born ensemble rather than only in a forced favorable branch; separately, a declared interval of feedback imperfection tests their stability.

b. Dependencies and run matrix. At $N_y = 32, 48, 64$, compare perfect correction, a declared imperfect-correction grid, and a forced postselected branch. Physical stochastic points use $S = 100$; a deterministic postselected branch is not padded with duplicate samples.

c. Initialization, saves, estimator, and controls. Use the same observable definitions as P1, S1, and T1. Store branch probabilities for postselection and never describe a zero-event record as a stochastic ensemble.

d. Acceptance and downgrade. The physical Born-ensemble wall claim requires the perfect-correction point only. Robustness is claimed solely over the tested imperfect interval. Put this comparison in the supplement unless it is indispensable to a mechanism. No legacy comparison is required.

C. Campaign M3: noise and the BPJ bulk transition

a. Claim and status. Status PLANNED. This campaign tests, but does not assume, bulk–boundary locking under the noise perturbation used for the BPJ domain-wall-free bulk transition.

b. Dependencies and run matrix. First run the domain-wall-free geometry on a dense noise grid at $N = 16, 20, 24, 28, 32$, $S = 100$. Locate the bulk transition using the topological marker, purity gap, and bulk tangent diagnostic. Only then scan the wall geometries across the same noise grid and the primary N_y sequence.

c. Initialization, saves, estimator, and controls. Use independent burn-ins at every noise value. Save all P1, S1, and T1 observables and the exact noise convention. Estimate bulk and boundary pseudocritical points independently in each bootstrap resample.

d. Acceptance gate and downgrade. Bulk–boundary locking is accepted only if extrapolated critical noise values have compatible joint confidence intervals. A preemptive boundary crossover or transition is explicitly allowed and must not be forced to coincide with the bulk point. This campaign belongs in the supplement unless it reveals a central new phenomenon. Legacy comparison: none unless a later path audit establishes an exactly matching BPJ observable and noise convention.

XV. FIGURE ARCHITECTURE

A. Main figures

a. Figure 1: inherited BPJ platform and validated wall implementation. (a) Vector schematic of the four overcomplete-Wannier occupation measurements and the outcome-conditioned ancillary fSWAP; (b) explicit controller-interface and support-terminated walls, two wall orientations, periodic direction, and flux seam; (c) representative prepared Chern profile; (d) topological and trivial finite-size indices; (e) transverse-converged comparison of wall implementations. The figure validates the inherited topological platform and fixes the two microscopic wall implementations on which the new universal claims are built.

b. Figure 2: universal state data. (a) entropy versus log chord with a $q = 1, 2, 3$ coefficient inset; (b) quantum interval variance and k , with $\delta_{SF}(N_y)$ and its zero target in the inset; (c) the unnormalized trajectory-averaged entanglement contour; (d) max-mix wall-on/wall-off purification; (e) transverse profile of the residual entropy or purity defect. The squared correlator, small- k structure factor details, and topological sweep move to the supplement. Exact-Hamiltonian calibration appears only as a visually distinct gray reference.

c. Figure 3: chirality in three clocks. (a) opposite modular drifts, with the modular clock marked in the panel; (b) total Born $\chi(k, \omega)$ on both walls, including a score/conditional decomposition inset; (c) frozen-record spectral flow with an integer histogram inset. These are exactly three primary panels. A compact three-clock key may sit above them, but is not a fourth result panel. The caption states what each route proves and explicitly denies a quantized transport claim.

d. Figure 4: record-only defect CFT. (a) record spacetime raster and the replica bridge from p_ξ to $\mathcal{Z}_n$; (b) defect self-information versus L^{-2} before and after the fixed subtraction; (c) record spacetime collapse and v_{rec} ; (d) charge-resolved tower; (e) independent state-versus-record universal-number summary. Panels (c)–(e) are shown as CFT data only after the small-system replica normalization has passed.

e. Figure 5: tangent edge channel and purification. (a) tangent-cocycle construction after physical burn-in and alignment; (b) wall/trivial gap scaling; (c) globally normalized low-mode localization; (d) max-mix purification collapse; (e) recordwise tangent bridge; optional inset comparing trajectory and mean covariance.

B. Supplement and schematic requirements

Supplement figures contain the exact benchmark contact sheet, transverse gate, burn-in diagnostics, fit-window sensitivity, update-order and n_{shell} controls, conditional and score response pieces, twist closure, tangent rank/null diagnostics, postselection, mean-channel comparisons, and noise transition. Every caption states S , the independent sampling unit, initialization, burn/observation windows, estimator order, uncertainty type, and fit window.

All conceptual schematics should be vector/TikZ or native illustration assets rather than screenshots. The key sketches are: the actual measurement-and-feedback tile; the two-wall cylinder with entanglement interval, response source, and twist seam; a taxonomy from one record to C_ξ , p_ξ , and $D\Phi_\xi$ versus $\bar{C}$; and the three-clock chirality diagram. A generic brick-wall circuit is not an adequate substitute for the implemented overlapping-orbital schedule.

XVI. PRODUCTION ORDER AND A100 RESOURCE STAGING

Execute in the following stop/go order:

1. V0 small-system validation.
2. V1 serial-schedule and order-artifact validation.
3. B0 exact benchmark and the optional B1 controller-frame pilot.
4. P1 bulk topology.
5. W1 transverse gate and selection of the actual production N_x .
6. Burn-in pilot of Appendix C.
7. S1 state CFT, including the matched entropy–variance spectral test, and S2 topological sweep.
8. T1 and P2, with tangent-frame rank pilots before full sizes.
9. B2 mechanism analysis, reusing the stationary activity records and the completed T1 and P2 outputs.
10. H1, H2, and H3 as three separate datasets.
11. R1 long record campaign.
12. M1–M3 controls.

Each stage begins with one parameter point and four samples, then a complete $S = 100$ pilot at the smallest accepted size, then the full family. Batch size is autotuned against the 40 GB A100 budget with at least 20% headroom, but a trajectory is never divided in a way that changes its ordered updates or random stream. Colab notebooks import the canonical GPU source as a separate file and end by disconnecting the runtime.

TABLE IV. Mandatory interpretation when a gate fails.

observation	licensed wording
c fit works but charge/correlator fails	entropy is consistent with a $c = 1$ wall; no full CFT identification
c and k fit separately but the matched $\pi^2/3$ relation fails	report separate entropy and current coefficients; no minimal $U(1)_1$ identification
wall tangent gap decreases but intercept unresolved	finite-size softening and wall-localized slow mode
modular drift only	handed modular flow, not physical chiral propagation
conditional response only	fixed-record response, not Born-ensemble response
χ chiral but twist not robust	chiral response without an extracted integer wall index
bulk Bott is quantized but $n_{\text{shell}} = 1$ marker does not converge	topological index with nonquantized local-marker finite-range corrections
replica mapping fails	report a defect self-information coefficient, not a record central charge
replica mapping passes but v_{rec} is unresolved	report $v_{\text{rec}} c_{\text{eff}}^{\text{rec}}$ only
one record gap only	slow record sector, not an operator spectrum
explicit-interface and support-terminated walls disagree	two distinct programmable interfaces
bulk and boundary noise points differ	preemptive boundary crossover/transition, not bulk–boundary locking

XVII. CLAIM-DOWNGRADE MATRIX

XVIII. DEFERRED WORK

The core campaign excludes a Chern number assigned to an ad hoc quenched-log tangent Hamiltonian, polar-unitary alignment of independently sampled states, a full scattering network, regularized-Choi endpoint spectroscopy, an absorbing-transition phase diagram, high-rank many-body reconstruction, a broad non-Hermitian survey, and a full Lindblad phase diagram. It also excludes promoting the natural-orbital coarse graining in `notebooks/flattened_hamiltonian_analysis/gaussian_entanglement_rg/` to a MERA result. That legacy analysis usefully tests whether locally mixed modes retained under quasi-one-dimensional Gaussian coarse graining move toward the walls, but its isometries are scale dependent, its repeated-tensor fits have order-one mismatch, and purified variants explicitly project the coarse covariance back to a Slater projector. It becomes main-text relevant only after a repeated finite-depth tensor has controlled discarded-mode error, stable state fidelity or correlators, and a nontrivial prediction beyond re-extracting $c \simeq 1$. These topics are follow-up projects unless a core gate specifically requires one.

Appendix A: Charge full counting statistics and the level-one current algebra

For a Gaussian trajectory with subsystem correlation matrix $C_{\xi,A}$, the quantum FCS of Q_A is

$$\mathcal{Z}_{\xi,A}(\lambda) = \det[\mathbf{1} - C_{\xi,A} + e^{i\lambda} C_{\xi,A}], \quad \mathcal{K}_{\xi,A}(\lambda) = \log \mathcal{Z}_{\xi,A}(\lambda). \quad (\text{A1})$$

With $\kappa_{n,\xi}^q = (-i\partial_\lambda)^n \mathcal{K}|_{\lambda=0}$,

$$\kappa_{1,\xi}^q = \text{Tr } C_{\xi,A}, \quad (\text{A2})$$

$$\kappa_{2,\xi}^q = \text{Tr}[C_{\xi,A}(\mathbf{1} - C_{\xi,A})], \quad (\text{A3})$$

$$\kappa_{3,\xi}^q = \text{Tr}[C_{\xi,A}(\mathbf{1} - C_{\xi,A})(\mathbf{1} - 2C_{\xi,A})], \quad (\text{A4})$$

$$\kappa_{4,\xi}^q = \text{Tr}[C_{\xi,A}(\mathbf{1} - C_{\xi,A})(\mathbf{1} - 6C_{\xi,A} + 6C_{\xi,A}^2)]. \quad (\text{A5})$$

The state-CFT observable is $\mathbb{E}[\kappa_{n,\xi}^q]$. It is distinct from the FCS of the Born mixture,

$$\mathcal{Z}_A^{\text{mix}}(\lambda) = \mathbb{E}_\xi[\mathcal{Z}_{\xi,A}(\lambda)], \quad (\text{A6})$$

whose logarithm includes classical trajectory fluctuations. In particular,

$$\text{Var}_{\text{mix}}(Q_A) = \mathbb{E}_\xi[\kappa_{2,\xi}^q] + \text{Var}_\xi(\kappa_{1,\xi}^q). \quad (\text{A7})$$

Both terms in Eq. (A7) are saved and plotted separately.

For a chiral $U(1)$ current, the normalization convention is fixed by

$$\begin{aligned} J(z)J(0) &\sim \frac{k}{z^2}, \\ [J_m, J_n] &= k m \delta_{m+n,0}. \end{aligned} \quad (\text{A8})$$

A free chiral complex fermion has $k = 1$. A single wall contributes

$$X_L(\ell) = \log \left[\frac{N_y}{\pi a} \sin \left(\frac{\pi \ell}{N_y} \right) \right], \quad \kappa_2^{(1w)}(\ell) = \frac{k}{2\pi^2} X_L(\ell) + b + \dots, \quad (\text{A9})$$

whereas two well-separated, oppositely oriented walls contribute

$$\kappa_2^{(2w)}(\ell) = \frac{k}{\pi^2} X_L(\ell) + b + \dots. \quad (\text{A10})$$

The common correlation-spectrum derivation of the matching entropy coefficients and the $\pi^2/3$ relation is given in Appendix B. The charged primaries obey $h_q = q^2/(2k)$, giving Eq. (49) at $k = 1$. The analysis must declare whether a subsystem contains one or two walls before converting a fit coefficient to k or c_{edge} . The coefficient k identifies the current algebra; by itself it does not determine chirality, which is tested only by Campaigns H1–H3.

Signed feedback-transfer FCS is a different classical observable. If $Y_e \in \{-1, 0, 1\}$ is the system-layer charge transfer at event e , save

$$Q_{\text{rec}}(R, T) = \sum_{e \in R \times T} Y_e, \quad \theta_R(s, T) = \frac{1}{T} \log \mathbb{E}_\xi[e^{-s Q_{\text{rec}}}]. \quad (\text{A11})$$

The empirical finite-time θ_R is not called a tilted-channel eigenvalue. Use log-sum-exp stabilization, trajectory bootstrap intervals, and a reweighting effective sample-size flag. Record FCS and quantum subsystem FCS may be compared only after their distinct definitions are displayed.

Appendix B: Common correlation-spectrum origin of entropy and charge fluctuations

This appendix fixes the exact Gaussian relation between interval entanglement and quantum charge fluctuations and then states the stronger CFT test used in Campaign S1. It applies to a number-conserving Gaussian reduced state whether or not the reduced density matrix is pure. It does not identify the within-trajectory quantum variance with classical trajectory-to-trajectory charge wandering.

1. Spectral calculus of the restricted covariance

Let

$$C_{\xi,A} u_{\xi a} = \nu_{\xi a} u_{\xi a}, \quad 0 \leq \nu_{\xi a} \leq 1, \quad (\text{B1})$$

and define the positive charge-susceptibility operator

$$\mathcal{V}_{\xi,A} = C_{\xi,A}(\mathbf{1} - C_{\xi,A}), \quad q_{\xi a} = \nu_{\xi a}(1 - \nu_{\xi a}) \in [0, 1/4]. \quad (\text{B2})$$

The interval quantum variance and von Neumann entropy are

$$F_{\xi,A}^q = \text{Tr } \mathcal{V}_{\xi,A} = \sum_a q_{\xi a}, \quad (\text{B3})$$

$$\begin{aligned} S_{\xi,A} &= -\text{Tr}[C_{\xi,A} \log C_{\xi,A} + (\mathbf{1} - C_{\xi,A}) \log(\mathbf{1} - C_{\xi,A})] \\ &= \sum_a h(\nu_{\xi a}) = \text{Tr } g(\mathcal{V}_{\xi,A}), \end{aligned} \quad (\text{B4})$$

where

$$h(\nu) = -\nu \log \nu - (1 - \nu) \log(1 - \nu), \quad g(q) = h\left(\frac{1 + \sqrt{1 - 4q}}{2}\right). \quad (\text{B5})$$

The branch of the square root is immaterial because $h(\nu) = h(1 - \nu)$. More generally,

$$S_{\xi,A}^{(n)} = \text{Tr } g_n(\mathcal{V}_{\xi,A}), \quad g_n(q) = \frac{1}{1 - n} \log[\nu(q)^n + (1 - \nu(q))^n], \quad (\text{B6})$$

with $\nu(q) = (1 + \sqrt{1 - 4q})/2$. Thus entropy and variance are different spectral functionals of the same operator. The scalar $F_{\xi,A}^q$ alone does not determine $S_{\xi,A}$: distinct multisets $\{q_{\xi a}\}$ can have the same first moment. The full spectrum of $\mathcal{V}_{\xi,A}$ does determine every Gaussian Rényi entropy.

The corresponding single-particle modular Hamiltonian is

$$h_{\xi,A}^{\text{mod}} = \log\left[(\mathbf{1} - C_{\xi,A})C_{\xi,A}^{-1}\right], \quad \nu_{\xi a} = \frac{1}{1 + e^{\epsilon_{\xi a}}}, \quad (\text{B7})$$

with zero and unit occupations understood by continuity. In modular-energy variables,

$$q(\epsilon) = \frac{1}{4 \cosh^2(\epsilon/2)}, \quad (\text{B8})$$

$$s(\epsilon) = \log\left[2 \cosh\left(\frac{\epsilon}{2}\right)\right] - \frac{\epsilon}{2} \tanh\left(\frac{\epsilon}{2}\right). \quad (\text{B9})$$

If $\rho_{\xi,A}(\epsilon) = \sum_a \delta(\epsilon - \epsilon_{\xi a})$, then

$$F_{\xi,A}^q = \int d\epsilon \rho_{\xi,A}(\epsilon) q(\epsilon), \quad S_{\xi,A} = \int d\epsilon \rho_{\xi,A}(\epsilon) s(\epsilon). \quad (\text{B10})$$

Both kernels select entanglement levels with $|\epsilon| = O(1)$, and

$$\int_{-\infty}^{\infty} d\epsilon q(\epsilon) = 1, \quad \int_{-\infty}^{\infty} d\epsilon s(\epsilon) = \frac{\pi^2}{3}. \quad (\text{B11})$$

Consequently a common slowly varying, logarithmically growing low-modular-energy density produces $S = (\pi^2/3)F^q + O(1)$. Equation (B11) explains the spectral origin of the coefficient but is not an exact finite-size equality for an arbitrary covariance spectrum.

2. CFT coefficient and level-one test

For the interval convention of Appendix A, one chiral wall contributes

$$S_n^{(1w)}(\ell) = \frac{c}{12} \left(1 + \frac{1}{n}\right) X_L(\ell) + b_{S_n} + \dots, \quad (\text{B12})$$

$$F_q^{(1w)}(\ell) = \frac{k}{2\pi^2} X_L(\ell) + b_F + \dots, \quad (\text{B13})$$

whereas two well-separated oppositely oriented walls contribute twice these logarithmic coefficients,

$$S_n^{(2w)}(\ell) = \frac{c}{6} \left(1 + \frac{1}{n}\right) X_L(\ell) + b'_{S_n} + \dots, \quad (\text{B14})$$

$$F_q^{(2w)}(\ell) = \frac{k}{\pi^2} X_L(\ell) + b'_F + \dots. \quad (\text{B15})$$

The geometry factor cancels in the ratio. Denoting the corresponding log-chord slopes by a_{S_n} and a_F gives

$$\frac{a_{S_n}}{a_F} = \frac{\pi^2 c}{6k} \left(1 + \frac{1}{n}\right), \quad \frac{a_S}{a_F} = \frac{\pi^2 c}{3k}. \quad (\text{B16})$$

For a charge-one chiral complex fermion, $c = k = 1$, and therefore

$$a_S - \frac{\pi^2}{3}a_F = 0, \quad a_{S_n} - \frac{\pi^2}{6} \left(1 + \frac{1}{n}\right) a_F = 0. \quad (\text{B17})$$

Equivalently,

$$S(\ell) - \frac{\pi^2}{3}F^q(\ell) = \text{constant} + \text{finite-size corrections} \quad (\text{B18})$$

has no logarithmic term. This is a leading-scaling statement, not a claim that the two spectral kernels agree mode by mode.

Under a unitary chiral-CFT interpretation and a fixed microscopic normalization of charge, the $U(1)$ current generates a $c = 1$ Sugawara sector. The combination $c \simeq 1$, $k \simeq 1$, Eq. (B17), and an independent H1–H3 chirality test supports a minimal $U(1)_1$ charged wall without an additional neutral gapless sector. Entropy and variance alone do not determine the sign of the chiral central charge or the direction of propagation.

3. Cross-boundary coherence and intrinsic mixedness

For a pure Slater trajectory the full correlation matrix is a projector. With the spatial block decomposition

$$C_\xi = \begin{pmatrix} C_{\xi,A} & C_{\xi,AB} \\ C_{\xi,BA} & C_{\xi,B} \end{pmatrix}, \quad C_\xi^2 = C_\xi, \quad (\text{B19})$$

the AA block of the projector identity gives

$$\mathcal{V}_{\xi,A} = C_{\xi,A}(\mathbf{1} - C_{\xi,A}) = C_{\xi,AB}C_{\xi,BA}. \quad (\text{B20})$$

Hence the $q_{\xi a}$ are the squared singular values of the cross-boundary correlation block,

$$F_{\xi,A}^q = \|C_{\xi,AB}\|_F^2, \quad S_{\xi,A} = \text{Tr } g(C_{\xi,AB}C_{\xi,BA}). \quad (\text{B21})$$

The variance is the total squared cross-boundary coherence, whereas the entropy retains the complete singular-value distribution.

For a globally mixed Gaussian state, $0 \leq C_\xi \leq \mathbf{1}$ but $C_\xi^2 \neq C_\xi$, and the exact decomposition becomes

$$\mathcal{V}_{\xi,A} = C_{\xi,AB}C_{\xi,BA} + (C_\xi - C_\xi^2)_{AA}. \quad (\text{B22})$$

Both terms are positive. The first measures cross-boundary coherence and the second is intrinsic mixedness already present in the global state. Their traces add in the charge variance, but the entropy is not generally the sum of separate functions of the two terms because they need not commute. Equation (B22) is the appropriate diagnostic during max-mix purification; Eq. (B20) is used only after the pure-trajectory gate passes.

The shared natural orbitals also define matched real-space contours,

$$f_{\xi,A}(i) = \sum_a |u_{\xi a}(i)|^2 q_{\xi a}, \quad s_{\xi,A}(i) = \sum_a |u_{\xi a}(i)|^2 h(\nu_{\xi a}), \quad (\text{B23})$$

with the exact sum rules

$$\sum_{i \in A} f_{\xi,A}(i) = F_{\xi,A}^q, \quad \sum_{i \in A} s_{\xi,A}(i) = S_{\xi,A}. \quad (\text{B24})$$

Comparing these unnormalized contours tests whether the same wall-localized natural orbitals carry both logarithms. Each contour is averaged over trajectories only after its per-trajectory construction.

4. Full charge statistics and ensemble decomposition

Writing $z = e^{i\lambda}$, the interval FCS polynomial factorizes as

$$\chi_{\xi,A}(z) = \det[\mathbf{1} - C_{\xi,A} + zC_{\xi,A}] = \prod_a (1 - \nu_{\xi a} + z\nu_{\xi a}). \quad (\text{B25})$$

Its zeros obey

$$z_{\xi a} = -\frac{1 - \nu_{\xi a}}{\nu_{\xi a}} = -e^{\epsilon_{\xi a}}. \quad (\text{B26})$$

Thus the exact full interval-charge distribution, whose probabilities are the coefficients of $\chi_{\xi,A}(z)$, determines the single-particle entanglement spectrum up to permutation and hence every Gaussian entropy. Numerical inversion of high-degree noisy counting polynomials is ill conditioned and is not a production gate; direct covariance eigenvalues remain the primary simulated estimator.

Finally, if one first samples a trajectory and then measures its interval charge, the law of total variance is

$$\text{Var}_{\text{total}}(N_A) = \mathbb{E}_{\xi}[\text{Tr } \mathcal{V}_{\xi,A}] + \text{Var}_{\xi}[\text{Tr } C_{\xi,A}]. \quad (\text{B27})$$

Only the first term is the quantum interval variance governed by the spectrum in Eq. (B2) and used to infer k . Boundary localization of the second term is meaningful dynamical information about the trajectory ensemble, but it is not substituted for the current-algebra coefficient.

5. Production estimator and acceptance rule

Campaign S1 computes $S_{\xi,A}$, $F_{\xi,A}^q$, their contours, and the $q_{\xi a}$ spectrum from the same restricted covariance for every declared interval. Origin and stationary-snapshot averages are internal to a trajectory and are performed identically for both observables. The common linear fit then produces $m_{1,\xi}$, f_{ξ} , and the paired difference $\delta_{SF,\xi}$ in Eq. (32). Whole trajectories are resampled jointly, preserving the covariance between the two slopes and among all interval lengths.

The primary test is the thermodynamic extrapolation of $\delta_{SF}^{\text{Born}}$ and the equivalent residual curve in Eq. (33). The ratio a_S/a_F and the Rényi relations in Eq. (34) are secondary. Repeat the complete paired analysis after each preregistered fit-window deletion and on the exact-Hamiltonian, matched trivial, and one-versus-two-wall controls. A nonzero residual logarithmic coefficient forbids the minimal $U(1)_1$ identification even if entropy and variance separately admit logarithmic fits. It does not erase the separate estimates of c and k , and it says nothing by itself about chirality.

a. Legacy comparison. No saved legacy artifact contains matched full-width interval entropy and within-trajectory interval variance on the same pure S1 covariance snapshots. The streaming pure-state campaign saves the entropy curves but not the required interval variance; the max-mix campaign saves total entropy and total charge variance during purification, which are different observables. Legacy data therefore guide sizes and fit windows only and do not supply a point in the $\pi^2/3$ test.

Appendix C: Circuit depth, Lu bound, wall hybridization, and burn-in

For a product input and a depth- D local circuit, the rank-entropy growth across region A obeys the Lu–Lessa–Kim–Hsieh bound [2]

$$S_0(A, T) - S_0(A, 0) \lesssim D |\partial A|. \quad (\text{C1})$$

For the full-width strip $A = [0, N_x) \times [0, \ell)$, $|\partial A| \sim 2N_x$. A two-wall CFT contribution therefore requires parametrically

$$\frac{1}{3} \log N_y \lesssim \kappa D N_x. \quad (\text{C2})$$

Here D counts layers of mutually nonoverlapping local maps, not circuit sweeps. The production random-serial schedule visits overlapping orbitals sequentially, so one cycle has a size-dependent implementation depth. For $n_{\text{shell}} = 1$, an update centered at $u = (x, y)$ has the periodic 3×3 cell support

$$\mathcal{S}_u = \{u + (\delta_x, \delta_y) : \delta_x, \delta_y \in \{-1, 0, 1\}\}. \quad (\text{C3})$$

The conflict graph joins u and v when $\mathcal{S}_u \cap \mathcal{S}_v \neq \emptyset$, equivalently when their cyclic separations obey

$$d_{N_x}(x, x') \leq 2, \quad d_{N_y}(y, y') \leq 2, \quad d_L(a, b) = \min_{m \in \mathbb{Z}} |a - b + mL|. \quad (\text{C4})$$

It has maximum degree at most 24 and hence at most 25 colors by a greedy bound. If $q_x \mid N_x$, $q_y \mid N_y$, and $q_x, q_y \geq 3$, the explicit residue coloring

$$c(x, y) = (x \bmod q_x, y \bmod q_y) \quad (\text{C5})$$

uses $q_x q_y$ colors and is proper: distinct equal-color centers have separation at least three in one coordinate. A commensurate 3×3 coloring therefore uses nine colors. For every size in Eq. (4), the conservative 4×4 coloring wraps cleanly and gives $\chi_{\text{conf}} \leq 16$.

Let $d_{\text{inst}} = O(1)$ denote the physical layer depth of one occupation-measurement-feedforward-fresh-ancilla-fSWAP instrument block. Keeping the four controller channels as separate blocks gives

$$d_{\text{cyc}} \leq 4\chi_{\text{conf}}d_{\text{inst}} \leq 64d_{\text{inst}}, \quad D = d_{\text{cyc}}T_{\text{cyc}}. \quad (\text{C6})$$

This is an $O(1)$ bound on the physical depth of one colored sweep, not on the total preparation depth. The BPJ bulk can converge in $O(1)$ sweeps while the additional critical wall requires a cycle count that grows with N_y ; the two statements concern different target entanglement structures. Equation (C2) gives only

$$T_{\text{cyc}} \gtrsim \frac{\log N_y}{3\kappa d_{\text{cyc}} N_x}. \quad (\text{C7})$$

The unknown coefficient makes this too weak to choose 10 versus 50 versus $2N_y$ cycles; in particular, it does not imply a $2N_y$ burn-in.

This coloring gives an optional quantum-simulation compilation. Within a color, disjoint local instruments commute, so simultaneous hardware readout and sequential numerical evaluation define the same joint Born instrument even though its outcomes need not be statistically independent. Reordering different color blocks is not a parallel reinterpretation of a saved random-serial trajectory because overlapping conditional maps generally do not commute. The production campaign therefore makes no constant-depth-per-sweep claim for its random-serial word; it reports Eq. (C6) only as a realizable resource bound for the corresponding color-block hardware schedule.

The circuit-specific operator-Schmidt-rank bounds are

$$r(a_\chi), r(a_\chi^\dagger) \leq 2, \quad r(n_\chi), r(\mathbf{1} - n_\chi) \leq 4. \quad (\text{C8})$$

Along record ξ , a sharper bookkeeping bound is

$$\Delta S_0^\xi \leq 2 \log 2 N_{=}^\partial(\xi) + \log 2 N_{\neq}^\partial(\xi), \quad (\text{C9})$$

with the two counts distinguished according to the boundary-crossing branch type. These bounds assume an initially product state. A random Slater determinant is already entangled and cannot be used to validate a product-input depth bound. The maximally mixed state is treated separately as a purification problem beginning at $t = 0$.

Finite transverse width produces an independent infrared scale. If the opposing-wall hybridization behaves as

$$m(w) \sim m_0 e^{-w/\xi_\perp}, \quad (\text{C10})$$

then one requires

$$m(w) \ll \frac{2\pi v}{N_y}, \quad w \gtrsim \xi_\perp \log N_y \quad (\text{C11})$$

up to nonuniversal constants. B0 determines $\xi_\perp$ and W1 verifies the same condition in the stochastic circuit.

The burn-in pilot runs $S = 100$ independent trajectories for at least $4N_y$ cycles with dense checkpoints. For each trajectory family define T_ϵ as the first cycle after which the bulk marker, entropy coefficient, charge statistics, wall profiles, purity diagnostics, and successive-covariance changes all remain within predeclared tolerances for a subsequent window of length N_y . Fit the ensemble distribution and upper confidence limit of $T_\epsilon(N_y)$ to constant, logarithmic, and power-law forms. The production burn-in is the upper 95% confidence bound, rounded up to a whole cycle, followed by a separate observation window.

The initial hypothesis is $2N_y$ preparation plus $2N_y$ observation. It is tested, not assumed. If stationarity fails, extend preparation in N_y increments up to $16N_y$. There are always 100 distinct burn-in histories; never burn in one trajectory and clone its state into 100 continuations.

Max-mix purification has no burn-in. Its clock starts at the initial $C = \mathbf{1}/2$. Tangent runs first complete the physical burn-in, then initialize the tangent frame, discard a declared alignment interval, reset only the QR log accumulator, and finally begin exponent accumulation. This procedure is tailored to the production serial implementation and must be re-evaluated if the actual overlap-conflict schedule changes.

Appendix D: Campaign-card checklist

Before a campaign is marked complete, its final note must answer all of the following:

1. Was the canonical GPU entry point recorded?
2. Were all 100 trajectories independently initialized and burned in where applicable?
3. Were max-mix and tangent clocks treated according to Appendix C?
4. Can every plotted scalar be reconstructed from per-trajectory raw data?
5. Is the order of nonlinear evaluation, time/source averaging, and trajectory averaging explicit?
6. Are fit windows, excluded points, and correction models preregistered and saved?
7. Do paired controls share declared random streams without being called the same record?
8. Did every shard pass manifest, seed, resume, and numerical-residual checks?
9. Does the caption state sample count, independent unit, uncertainty, and geometry?
10. If the gate failed, was the prescribed downgrade used without changing the estimator?

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- [1] A. Bhuiyan, H. Pan, and C.-M. Jian, [Physical Review Research](#) **8**, 023147 (2026), [arXiv:2507.13437 \[quant-ph\]](#).
 [2] T.-C. Lu, L. A. Lessa, I. H. Kim, and T. H. Hsieh, [PRX Quantum](#) **3**, 040337 (2022).